Dirac Operators on Loop Spaces

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Gregory D. Landweber

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Abstract

In his paper "S^1 Actions and Elliptic Genera", Taubes computes the index of the Dirac operator on a free loop space, reducing the problem to finding the index in an infinitesimal neighborhood of the constant loops, using localization with respect to the circle action rotating the loops. This argument requires only a local Dirac operator on the normal bundle to the constant loops, while the global Dirac operator itself is considered only formally. This thesis gives an explicit construction for that global Dirac operator in the case of loop groups and homogeneous loop spaces, substituting the representation theory of loop groups and their Lie algebras for the otherwise intractable infinite dimensional analysis. The global case introduces an additional parameter, the central charge, and Taubes' local version of the Dirac operator emerges as the infinite central charge limit of the global Dirac operator.

In the first half of this thesis, we discuss the Dirac operator on compact Lie groups and their homogeneous spaces. Noting that the ubiquitous constant ρ appearing in the Weyl character formula and the Borel-Weil-Bott theorem is in fact the highest weight of the spin representation, many of the fundamental results in the representation theory of Lie groups have a natural interpretation in terms of spinors and the Dirac operator. In the second half, we discuss the analogues of these results in the representation theory of loop groups. The primary complication in the loop group picture is determining the proper domain on which the Dirac operator acts. Using a Hilbert space of sections modeled on the Peter-Weyl decomposition of $L^2(G)$, we analyze the Dirac operator on a homogeneous loop space by "looping" the techniques in Bott's paper "The index theorem for homogeneous differential operators". In the process, we examine the representation ring of a loop group, and we prove a homogeneous version of the Weyl-Kac character formula—based on a similar result for compact Lie groups recently noted by Gross, Kostant, Ramond, and Sternberg—in order to compute the index of the Dirac operator.

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Introduction

The goal of this thesis is to provide an explicit construction for the Dirac operator on certain free loop spaces. Although the motivation for such a Dirac operator originated in string theory, it has been shown to have topological applications involving elliptic genera. The index of this Dirac operator has already been computed in the literature. In [**7**], Taubes shows that localization with respect to the circle action given by rotation of loops reduces the problem to finding the index in an infinitesimal neighborhood of the constant loops. Using techniques from quantum field theory, it is then possible to construct a local Dirac operator and compute its index. However, a global version of the Dirac operator itself has thus far proved elusive.

Our approach is to construct the Dirac operator for loop groups and for loops on homogeneous spaces. This allows us to substitute an algebraic problem, now involving the representation theory of loop groups and infinite dimensional Lie algebras, for the otherwise intractable analysis on infinite dimensional topological spaces. The Dirac operator on homogeneous spaces G/H plays a fundamental role in the representation theory of Lie groups. The ubiquitous ρ-shift appearing in the Weyl character formula and the Borel-Weil-Bott theorem can be interpreted in terms of spinors and the Dirac operator on G/T, where T is a maximal torus of G. There are also analogous results for G/H when H is of maximal rank in G (see [**2**] and [**3**]). Many of these results carry over directly to the loop group case. As described in [**6**], positive energy representations of a loop group LG can be understood via Borel-Weil theory on the homogeneous space $(LG)/T$, and the Weyl-Kac character formula again involves the spinors on $(LG)/T$, given by the spinors on G/T tensored with the exterior algebra on the positive energy portion of the Lie algebra $L\mathfrak{g}_{\mathbb{C}}$. This spin representation can be thought of either as a Fermionic Fock space or as a positive energy representation of LG of level (central charge) $c = \alpha_{\max} \cdot \rho + 1$, where $\alpha_{\max}$ is the highest root of G, and the basic inner product satisfies $\|\alpha_{\max}\|^2 = 2$. So the ρ-shift for Lie groups now becomes a (ρ, c)-shift for loop groups.

The Dirac operator for homogeneous loop spaces can be written formally as a generalization of the Dirac operator on G/H, which is examined by Kostant in [**4**]. In the loop group case, the difficulty lies in finding an appropriate

domain for the Dirac operator. Although there is no immediate generalization of the Peter-Weyl theorem for loop groups, we can still use it as our model for the space of spinors on LG, restricting to the Hilbert space $\widehat{\bigoplus}\mathcal{H}_{\lambda,h} \otimes \mathcal{H}^*_{\lambda,h} \otimes \mathcal{S}^*$. (Here we take the negative energy spin representation $\mathcal{S}^*$ of level $-c$, dual to the positive energy spin representation appearing in the Weyl-Kac character formula.) This allows us to examine the Dirac operator one positive energy representation at a time. In [**8**], Teleman takes this approach to analyze the related Dolbeault complex on LG/G, which he interprets in terms of the relative Lie algebra cohomology $H^*(\mathfrak{g}_{\mathbb{C}[z]}, \mathfrak{g}_{\mathbb{C}}; \mathcal{H}^*_{\lambda,h})$ with coefficients in each of the irreducible negative energy representations of LG.

We are primarily interested in the Dirac operator on the homogeneous loop space $L(G/H) = LG/LH$. When considering LG/LH, we still have complete reducibility for representations of LH, but those representations are no longer finite dimensional, except in the case where the level h vanishes and the representation is trivial. Our model for the spinors on LG/LH is now

$$\widehat{\bigoplus}_{\lambda,h} \mathcal{H}_{\lambda,h} \otimes \left(\mathcal{H}^*_{\lambda,h} \otimes \mathcal{S}^*\right)^{LH}.$$

Here the spin representation $\mathcal{S}^*$ is a negative energy representation of LH of level $c = -c_G + c_H$, and the product $\mathcal{H}^*_{\lambda,h} \otimes \mathcal{S}^*$ has nontrivial level $-h - c < 0$. Thus, there is no LH-invariant subspace, and the proposed domain of our Dirac operator is empty! To bypass this problem, we consider the Dirac operator twisted by a homogeneous bundle over LG/LH associated to a positive energy representation $\mathcal{U}$ of LH of level $k \geq c$. The domain of this new operator is the space of sections of the bundle associated to the LH representation $\mathcal{S}^* \otimes \mathcal{U}$. Although this representation has positive level $k - c$, it is no longer a positive (or negative) energy representation.

To make sense of such mixed products of positive and negative energy representations, we develop some new machinery. We first define the representation ring $R(LG)$ to be the ring of virtual positive energy representations of LG. We also define $\hat{R}(LG)$ to be the group of formal infinite sums of irreducible positive energy representations, where the minimal energies of the summands are bounded above. Then, in analogy with [**2**], we construct an induction map $i_* : \hat{R}(LH) \to \hat{R}(LG)$, formally adjoint to the restriction homomorphism $i^* : R(LG) \to R(LH)$. Using this machinery, we show that $\mathcal{S} \otimes \mathcal{U}$ lives in $\hat{R}(LH)$, and that its image $i_*(\mathcal{S} \otimes \mathcal{U})$ under the induction map gives us a model for the domain of the Dirac operator on LG/LH twisted by $\mathcal{U}$. Furthermore, dividing the spin representation into half-spin representations $\mathcal{S}^+$ and $\mathcal{S}^-$, we show that the index of the Dirac operator is given by $i_*\left((\mathcal{S}^+ - \mathcal{S}^-)^* \otimes \mathcal{U}\right)$, where this difference is a finite element of $R(LG)$.

CHAPTER 1

Lie Groups

1.1. Homogeneous Spaces

Let G be a compact, connected Lie group, and let H be a closed connected subgroup of G. We are interested in the homogeneous space $M = G/H$ of right cosets. There is a natural G-action on M inherited from left multiplication on G, and H is the stabilizer of the identity coset eH. Indeed, given any manifold M with base point, where G acts transitively and H is the stabilizer of the base point, we find that M is equivariantly isomorphic to the coset space G/H.

A *homogeneous vector bundle* over M is a vector bundle $E \to M$ together with a lifting of the G-action on the base M to a fiberwise linear G-action on the total space E. All homogeneous vector bundles on M are induced by representations of H. Viewing $\pi : G \to M$ as a principal H-bundle, for any H-representation V we can consider the associated bundle $G \times_H V \to M$ given by

$$G \times_H V = G \times V/\{(g, v) \sim (gh, h^{-1}v)\}.$$

In other words, we take the quotient of $G \times V$ by the diagonal action of H. The G-action on $G \times_H V$ is just left multiplication on the G component, and the vector bundle projection takes $(g, v) \mapsto gH$. Conversely, given any homogeneous vector bundle $E \to M$, we can take $V = E_{eH}$ to be the fiber at the identity coset. Since H fixes eH, the G-action on E restricts to an H-action on V, and we find that E is isomorphic to the induced bundle $G \times_H V$.

Functions on the homogeneous space M correspond to right H-invariant functions on G, i.e., functions satisfying $f(gh) = f(g)$ for all $g \in G, h \in H$. On the other hand, sections of the vector bundle $G \times_H V$ correspond to right H-equivariant maps $s : G \to V$ satisfying $s(gh) = h^{-1}s(g)$, where the associated section is given by $s(gH) = (g, s(g)) \in (G \times_H V)_{gH}$. The H-equivariance ensures that the section is well-defined on M.

In particular, the tangent bundle TM and vector fields can be treated using the above methods. The tangent space at the identity coset is $TM_{eH} = \mathfrak{g}/\mathfrak{h}$. Given an inner product $\langle\ ,\ \rangle$ on $\mathfrak{g}$ invariant under the adjoint action of G (for now, we can use the Killing form, although later when we consider loop groups, we will instead use the basic inner product, rescaled so that the longest root has squared length 2), we may take an orthogonal decomposition $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}$ and

identify $TM_{eH} \cong \mathfrak{p}$. At the identity coset, H acts on $\mathfrak{p}$ by the adjoint action (the inner product is Ad-invariant, so $\operatorname{Ad} H$ fixes both $\mathfrak{h}$ and $\mathfrak{p}$). The tangent space is therefore isomorphic to the associated vector bundle $TM \cong G \times_H \mathfrak{p}$, and vector fields on M correspond to maps $\xi : G \to \mathfrak{p}$ satisfying the H-equivariance condition $\xi(gh) = (\operatorname{Ad} h^{-1})\xi(g)$.

The geometry of the homogeneous space M is induced by the geometry of G, so we now recall some of the basic geometry of Lie groups. Since G is compact it admits a bi-invariant metric, which is completely determined by the Ad-invariant inner product on $\mathfrak{g}$, the tangent space at the identity. Given left-invariant vector fields X_L, Y_L taking values $X, Y \in \mathfrak{g}$ at the identity, their bracket is again left-invariant and is given by $[X_L, Y_L] = [X, Y]_L$. Indeed this is usually taken as the the definition of the bracket on the Lie algebra $\mathfrak{g}$. For left-invariant vector fields, the Levi-Civita connection is given by $\nabla_{X_L} Y_L = \frac{1}{2}[X, Y]_L$, which is then torsion-free as we have

$$\nabla_{X_L} Y_L - \nabla_{Y_L} X_L = [X, Y]_L = [X_L, Y_L].$$

Trivializing the tangent space TG by left translation to agree with our conventions above, we may write vector fields on G as maps $G \to \mathfrak{g}$. The left-invariant vector fields then correspond to the constant maps $X_L(g) \equiv X$. Given two general vector fields $\xi, \eta : G \to \mathfrak{g}$, the Levi-Civita connection is then given by

$$(\nabla_\xi \eta)(g) = (\partial_\xi \eta)(g) + \frac{1}{2}\left[\xi(g), \eta(g)\right],$$

where ∂_ξ denotes the directional derivative for functions on G, and the bracket on the right hand side is the Lie algebra bracket on $\mathfrak{g}$.

Let $\pi : G \to M$ be the coset map. The bi-invariant metric on G then lets us identify $\pi^* TM$ with a subbundle of TG, so for any vector field ξ on M, the pullback $\pi^*\xi$ gives a lifting of ξ to a vector field on G. Conversely, given a vector field $\tilde{\xi}$ on G, its push forward $\pi_*\tilde{\xi}$ gives a well-defined vector field on M provided that $\tilde{\xi}$ is right H-equivariant. This machinery lets us define the geometry on M induced by the geometry on G. The metric on M is given by

$$\langle \xi, \eta \rangle_M = \langle \pi^*\xi, \pi^*\eta \rangle_G,$$

and the vector field bracket on M is

$$[\xi, \eta]_M = \pi_*[\pi^*\xi, \pi^*\eta]_G$$

(To be sure that this bracket is well-defined on M, one must first verify that $[\pi^*\xi, \pi^*\eta]_G$ is indeed right H-equivariant). The Levi-Civita connection on M is then given by

$$\nabla^M_\xi \eta = \pi_* \nabla^G_{\pi^*\xi} \pi^*\eta.$$

Writing vector fields on G and M as maps from G to $\mathfrak{g}$ and $\mathfrak{p}$ respectively, let $i : \mathfrak{p} \to \mathfrak{g}$ denote the inclusion and $p : \mathfrak{g} \to \mathfrak{p}$ denote the orthogonal projection.

We then have $(\pi^*\xi)(g) = (i \circ \xi)(g)$ and $(\pi_*\tilde{\xi})(g) = (p \circ \tilde{\xi})(g)$. Given vector fields $\xi, \eta : G \to \mathfrak{p}$ on M, the Levi-Civita connection then becomes

$$(\nabla^M_\xi \eta)(g) = (\partial_\xi \eta)(g) + \frac{1}{2}\left[\xi(g), \eta(g)\right]_{\mathfrak{p}}, \tag{1.1}$$

where $[\ ,\ ]_{\mathfrak{p}}$ denotes the Lie algebra bracket on $\mathfrak{g}$ projected onto $\mathfrak{p}$.

Example (Symmetric Spaces). At first glance, the expression (1.1) for the Levi-Civita connection looks quite straightforward, particularly if M is a symmetric space where $[\mathfrak{p}, \mathfrak{p}] \subset \mathfrak{h}$. In that case, (1.1) further simplifies to $(\nabla^M_\xi \eta)(g) = (\partial_{\xi(g)}\eta)(g)$. However, there is a hidden complication. This formula is written in terms of the trivialization of TG given by left translation, while the vector fields on M correspond to right H-equivariant vector fields on G. In particular, the left-invariant vector fields X_L on G do not descend to vector fields on M unless G is abelian. The simplest examples of right H-equivariant vector fields on G are the right invariant vector fields X_R of G, given by the maps $X_R : g \mapsto (\operatorname{Ad} g^{-1})X$. If M is a symmetric space, the covariant derivative of the corresponding vector field $\pi_* X_R$ on M with respect to the vector $(g, Y) \in G \times_H \mathfrak{p}$ is then given by

$$\begin{aligned}(\nabla^M_Y \pi_* X_R)(g) &= \frac{d}{dt}\left[p \circ \operatorname{Ad}(ge^{tY})^{-1}(X)\right]_{t=0} \\ &= -[Y, X_R(g)]_{\mathfrak{p}} \\ &= -[Y, X_R(g)|_{\mathfrak{h}}],\end{aligned}$$

where $|_{\mathfrak{h}}$ indicates the orthogonal projection onto $\mathfrak{h}$.

1.2. The Dirac Operator on G

We begin this section by reviewing the definition of a spin manifold. Given a vector space V with a symmetric bilinear form b, the *Clifford algebra* $\operatorname{Cl}(V)$ is the algebra generated by V subject to the relations $u \cdot v + v \cdot u = -2b(u, v)$. (Note that the opposite sign convention, taking $+2b(u, v)$ on the right hand side, also appears in the literature.) In our case, we let b be a positive definite inner product on V. Given an orthonormal basis e_i for V, we then have $e_i^2 = -1$ and $e_i \cdot e_j = -e_j \cdot e_i$ for $i \neq j$. The quadratic elements $e_i \cdot e_j$ for $i \neq j$ span the Lie algebra $\mathfrak{spin}(V)$, and we obtain the corresponding spin group $\operatorname{Spin}(V)$ by exponentiating within the Clifford algebra. Embedding V in $\operatorname{Cl}(V)$ as the linear elements, the spin group $\operatorname{Spin}(V)$ acts on V via the adjoint action $\operatorname{Ad}\omega : v \mapsto \omega \cdot v \cdot \omega^{-1}$ for $\omega \in \operatorname{Spin}(V)$ and $v \in V$. The map $\operatorname{Ad} : \operatorname{Spin}(V) \to \operatorname{SO}(V)$ is then a non-trivial double cover if $\dim V \geq 2$, and in fact $\operatorname{Spin}(n) = \operatorname{Spin}(\mathbb{R}^n)$ is the universal covering space of $\operatorname{SO}(n)$ for $n \geq 3$. The induced map on the Lie algebras is then an isomorphism $\operatorname{ad} : \mathfrak{spin}(V) \to \mathfrak{so}(V)$, and its inverse is given by the following lemma:

LEMMA 1. *Given a real vector space V with an inner product and an element $A \in \mathfrak{so}(V)$, the unique element $\omega_A \in \mathfrak{spin}(V)$ satisfying* $\operatorname{ad}\omega_A = A$ *is*

$$\omega_A = \frac{1}{4}\sum_i X_i^* \cdot A(X_i), \tag{1.2}$$

where $\{X_i\}$ is a basis for V and $\{X_i^*\}$ is the dual basis satisfying $\langle X_i, X_j^*\rangle = \delta_{ij}$.

PROOF. For any $v \in V$, and ω_A given by (1.2), we have

$$\begin{aligned}
\operatorname{ad}\omega_A : v \mapsto \frac{1}{4}\sum_i \big(X_i^* \cdot A(X_i)\cdot v - v \cdot X_i^* \cdot A(X_i)\big) \\
&= \frac{1}{4}\sum_i \big(-2X_i^*\langle A(X_i), v\rangle + 2\langle X_i^*, v\rangle A(X_i)\big) \\
&= \frac{1}{2}\sum_i \langle X_i, A(v)\rangle X_i^* + \frac{1}{2}A\Big(\sum_i \langle X_i^*, v\rangle X_i\Big) \\
&= A(v).
\end{aligned}$$

By a dimension count, we see that ω_A is uniquely determined by A. □

If $n = 2m$ is even, then we have $\mathrm{Cl}(\mathbb{C}^n) \cong \mathrm{Cl}(\mathbb{R}^n) \otimes \mathbb{C} \cong \mathrm{End}(\mathbb{S})$, where the *spin representation* $\mathbb{S}$ is a complex 2^m dimensional vector space. The group $\mathrm{Spin}(n) \subset \mathrm{Cl}(\mathbb{C}^n)$ acts on $\mathbb{S}$, which further decomposes into two distinct irreducible *half-spin representations* $\mathbb{S}^\pm$ of $\mathrm{Spin}(n)$. If $n = 2m+1$ is odd, then instead we have $\mathrm{Cl}(\mathbb{C}^n) \cong \mathrm{End}(\mathbb{S}^+) \oplus \mathrm{End}(\mathbb{S}^-)$, where $\dim_{\mathbb{C}} \mathbb{S}^\pm = 2^m$.

A Riemannian manifold M of dimension n is called a *spin manifold* if its principal $\mathrm{SO}(n)$ frame bundle P admits a double cover $Q \to P$ which restricts to the map $\mathrm{Ad} : \mathrm{Spin}(n) \to \mathrm{SO}(n)$ on each fiber. The choice of a principal $\mathrm{Spin}(n)$-bundle Q on M is called a *spin structure*. Given a spin structure on M, the tangent bundle is then the bundle $TM \cong Q \times_{\mathrm{Spin}(n)} \mathbb{R}^n$ associated to the double cover $\mathrm{Ad} : \mathrm{Spin}(n) \to \mathrm{SO}(n)$. We also have the spin bundle $S \cong Q \times_{\mathrm{Spin}(n)} \mathbb{S}$, associated to the spin representation, as well as the two half-spin bundles $S^\pm \cong Q \times_{\mathrm{Spin}(n)} \mathbb{S}^\pm$.

A Lie group G is automatically a spin manifold; since its tangent bundle can be trivialized, all of its characteristic classes, and hence any potential obstructions, must vanish. In terms of Lie groups, this corresponds to the fact from the theory of covering spaces that the adjoint representation $\mathrm{Ad} : G \to \mathrm{SO}(\mathfrak{g})$ can be lifted to a spin representation $\widetilde{\mathrm{Ad}} : \tilde{G} \to \mathrm{Spin}(\mathfrak{g})$, where $\tilde{G}$ is a finite cover of G such that $\pi_1(\tilde{G})$ is torsion free. (If G is semi-simple, we can let $\tilde{G}$ be the universal cover of G, or we can just require that G be simply connected from the start.) At the Lie algebra level, the choice of covering space makes no difference, and we see by applying (1.2) that the spin lift of

the adjoint representation $\mathrm{ad} : \mathfrak{g} \to \mathfrak{so}(\mathfrak{g})$ is the representation $\widetilde{\mathrm{ad}} : \mathfrak{g} \to \mathfrak{spin}(\mathfrak{g})$ given by

$$\widetilde{\mathrm{ad}}\, X = \frac{1}{4}\sum\nolimits_i X_i^* \cdot [X, X_i]. \tag{1.3}$$

To facilitate our computations below, we summarize here the various commutators that occur.

Corollary 1. *For any $X, Y \in \mathfrak{g}$, the following three commutation (and anti-commutation) identities hold in the Clifford algebra* $\mathrm{Cl}(\mathfrak{g})$*:*

1. $\{X, Y\} := X \cdot Y + Y \cdot X = -2\langle X, Y\rangle$,
2. $[\widetilde{\mathrm{ad}}\, X, \widetilde{\mathrm{ad}}\, Y] = \widetilde{\mathrm{ad}}[X, Y]$,
3. $[\widetilde{\mathrm{ad}}\, X, Y] = [X, Y]$,

where the bracket on the right side of (2) and (3) is the bracket in $\mathfrak{g}$.

Proof. The first identity is just the definition of the Clifford algebra, while the third follows immediately from Lemma 1 and our definition $\widetilde{\mathrm{ad}}\, X = \omega_{\mathrm{ad}\, X}$. For the second identity, noting that the map $A \mapsto \omega_A$ given by (1.2) is a Lie algebra isomorphism, we have $[\omega_{\mathrm{ad}\, X}, \omega_{\mathrm{ad}\, Y}] = \omega_{[\mathrm{ad}\, X, \mathrm{ad}\, Y]} = \omega_{\mathrm{ad}[X,Y]}$. □

On the tangent bundle TG, recall that the Levi-Civita connection is given by $\nabla_\xi \eta = \partial_\xi \eta + \frac{1}{2}\,\mathrm{ad}\,\xi(\eta)$, where we trivialize TG by left translation. Lifting to the spin bundle S, the Levi-Civita connection applied to a spinor $s \in \Gamma(S)$ is $\nabla_\xi s = \partial_\xi s + \frac{1}{2}\,\widetilde{\mathrm{ad}}\,\xi(s)$, where $\widetilde{\mathrm{ad}}\,\xi \in \mathfrak{spin}(\mathfrak{g})$ acts by the spin representation.

The *Dirac operator* on a spin manifold M is defined as

$$\not{\partial} : \Gamma(S) \xrightarrow{\nabla} \Gamma(T^*M \otimes S) \xrightarrow{\langle\,,\rangle} \Gamma(TM \otimes S) \xrightarrow{c} \Gamma(S),$$

the composition of the Clifford action c with the connection, using the metric to identify the tangent and cotangent spaces. With respect to a local frame, the Dirac operator is $\not{\partial} = \sum_i c(e_i^*)\nabla_{e_i}$. On a Lie group G, we can write the Dirac operator in terms of a basis of left-invariant vector fields, noting that

$$(\partial_{X_L} f)(g) = \frac{d}{dt}\, f(ge^{tX})\big|_{t=0} = \frac{d}{dt}\, r(e^{tX})f\big|_{t=0}(g) = (r(X)f)(g),$$

where $r(g)$ is the right action of G on functions, and $r(X)$ is the corresponding induced action of $\mathfrak{g}$. The Dirac operator on G then becomes

$$\not{\partial} = \sum\nolimits_i \Big(r(X_i) \otimes X_i^* + 1 \otimes \frac{1}{2} X_i^* \cdot \widetilde{\mathrm{ad}}\, X_i \Big). \tag{1.4}$$

Furthermore, we can view $\not{\partial}$ as a purely algebraic object living in the non-abelian Weil algebra $U(\mathfrak{g}) \otimes \mathrm{Cl}(\mathfrak{g})$. (See [**1**] for a discussion of this.) The cubic term in the Dirac operator can also be written as $\frac{1}{8}\sum_{i,j} X_i^* \cdot X_j^* \cdot [X_i, X_j]$, which is, up to a constant, the Clifford algebra representative of the closed fundamental 3-form ω on G given by $\omega(X, Y, Z) = \langle [X, Y], Z\rangle$.

The following technical lemma greatly simplifies computations involving the Dirac operator, letting us treat certain bilinear maps as if they are Ad-invariant, provided that we sum over a complete basis for $\mathfrak{g}$.

LEMMA 2. *If $\{X_i\}$ is a basis for $\mathfrak{g}$ and $\{X_i^*\}$ is the dual basis with respect to an Ad-invariant inner product on $\mathfrak{g}$, the identity*

$$\sum_i b\big([X_i, Y], X_i^*\big) = \sum_i b\big(X_i, [Y, X_i^*]\big)$$

holds for any bilinear map $b : \mathfrak{g} \times \mathfrak{g} \to V$ taking values in a vector space V.

PROOF. To move the commutator from one side to the other, we decompose it in terms of the basis, giving us

$$\begin{aligned}\sum_i b\big([X_i, Y], X_i^*\big) &= \sum_{i,j} b\left(\langle [X_i, Y], X_j^*\rangle X_j, X_i^*\right) \\ &= \sum_{i,j} b\left(X_j, \langle X_i, [Y, X_j^*]\rangle X_i^*\right) \\ &= \sum_j b\left(X_j, [Y, X_j^*]\right),\end{aligned}$$

where we use the Ad-invariance of the metric in the second line. □

For functions on G, recall that we have the left and right G-actions

$$l(h)f : g \mapsto f(h^{-1}g), \qquad r(h)f : g \mapsto f(gh)$$

respectively. For vector fields, we first introduce the notation gX and Xg to denote the left and right translates of a vector X. The left and right actions on vector fields are then given by

$$(l(h)\xi)_g = h\xi_{h^{-1}g}, \qquad (r(h)\xi)_g = \xi_{gh}h^{-1}$$

respectively. Trivializing the tangent bundle by left translation, we identify functions $\xi : G \to \mathfrak{g}$ with the vector fields given by $\xi_g = g\xi(g)$. In terms of such functions, the left and right actions on vector fields are now

$$l(h)\xi : g \mapsto \xi(h^{-1}g), \qquad r(h)\xi : g \mapsto h\xi(gh)h^{-1}.$$

So, while the left action on vector fields remains the same as the left action on the corresponding functions, the right action on vector fields becomes $\mathrm{Ad} \circ r$. Lifting to the spin bundle, we write the right action on spinors as ϱ to distinguish it from the right action r on functions, and we have $\varrho(h) = \widetilde{\mathrm{Ad}}(h) \circ r(h)$. (Note that ϱ is not necessarily a G-action, but rather an action for the cover $\tilde{G}$ of G discussed above.) The induced action on the Lie algebra $\mathfrak{g}$ is then $\varrho(Y) = r(Y) \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\, Y$. Since we wrote the Dirac operator in (1.4) entirely in terms of the right action r and the Clifford action, we conclude that the

Dirac operator must automatically commute with the left action on spinors. The following lemma then shows that the Dirac operator also commutes with the right action ϱ on spinors.

LEMMA 3. *The Dirac operator $\not{\partial}$ on G commutes with the elements*

$$\varrho(Y) = r(Y) \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\, Y \in U(\mathfrak{g}) \otimes \mathrm{Cl}(\mathfrak{g})$$

for all $Y \in \mathfrak{g}$.

PROOF. We compute the commutator in two parts. First, we consider the commutator with the $\sum_i r(X_i) \otimes X_i^*$ portion of the Dirac operator:

$$\begin{aligned}\Big[\varrho(Y), \sum_i r(X_i) \otimes X_i^*\Big] &= \sum_i \big([r(Y), r(X_i)] \otimes X_i^* + r(X_i) \otimes [\widetilde{\mathrm{ad}}\, Y, X_i^*]\big) \\ &= \sum_i \big(r([Y, X_i]) \otimes X_i^* + r(X_i) \otimes [Y, X_i^*]\big) \\ &= 0,\end{aligned}$$

in the final line applying Lemma 2 to the bilinear map $b(X, Y) = r(X) \otimes Y$. Next, we consider the commutator with the cubic term in the Dirac operator. Recalling the identity $[A, BC] = [A, B]C + B[A, C]$, we have

$$\begin{aligned}\Big[\widetilde{\mathrm{ad}}\, Y, \sum_i X_i^* \cdot \widetilde{\mathrm{ad}}\, X_i\Big] &= \sum_i \big[\widetilde{\mathrm{ad}}\, Y, X_i^* \cdot \widetilde{\mathrm{ad}}\, X_i\big] \\ &= \sum_i \big([\widetilde{\mathrm{ad}}\, Y, X_i^*] \cdot \widetilde{\mathrm{ad}}\, X_i + X_i^* \cdot [\widetilde{\mathrm{ad}}\, Y, \widetilde{\mathrm{ad}}\, X_i]\big) \\ &= \sum_i \big([Y, X_i^*] \cdot \widetilde{\mathrm{ad}}\, X_i + X_i^* \cdot \widetilde{\mathrm{ad}}[Y, X_i]\big) \\ &= 0,\end{aligned}$$

where we apply Lemma 2 to the bilinear map $b(X, Y) = X \cdot \widetilde{\mathrm{ad}}\, Y$. Combining these computations, we obtain the desired commutator $[\varrho(Y), \not{\partial}] \equiv 0$. □

We now compute the square of the Dirac operator. To simplify matters, we work with an orthonormal basis $\{X_i\}$ of $\mathfrak{g}$ where $X_i^* = X_i$. Breaking the Dirac operator into its two pieces, we first compute

$$\begin{aligned}\Big(\sum_i r(X_i) \otimes X_i\Big)^2 &= \sum_i r(X_i)^2 \otimes X_i^2 + \frac{1}{2}\sum_{i,j} r([X_i, X_j]) \otimes X_i \cdot X_j \\ &= -\sum_i r(X_i)^2 \otimes 1 + \frac{1}{2}\sum_{i,j} r(X_i) \otimes [X_j, X_i] \cdot X_j \\ &= -\sum_i r(X_i)^2 \otimes 1 + 2\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}\, X_i.\end{aligned}$$

In the first line, we break up the sum into its trace and skew-symmetric parts. Then we apply Lemma 2 to the bilinear map $b(X,Y) = r(X) \otimes Y$ and recognize $\widetilde{\mathrm{ad}}\, X_i$ in the result. Next, we use the identity $\{A, BC\} = \{A, B\}C - B[A, C]$ to compute the cross terms of the product:

$$\begin{aligned}
\frac{1}{2}\Big\{\sum_i r(X_i) \otimes X_i, 1 \otimes \sum_j X_j \cdot \widetilde{\mathrm{ad}}\, X_j\Big\} &= \frac{1}{2}\sum_{i,j} r(X_i) \otimes \{X_i, X_j \cdot \widetilde{\mathrm{ad}}\, X_j\} \\
&= \frac{1}{2}\sum_{i,j} r(X_i) \otimes \big(\{X_i, X_j\} \cdot \widetilde{\mathrm{ad}}\, X_j - X_j \cdot [X_i, X_j]\big) \\
&= \frac{1}{2}\sum_i r(X_i) \otimes (-2\,\widetilde{\mathrm{ad}}\, X_i - 4\,\widetilde{\mathrm{ad}}\, X_i) \\
&= -3\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}\, X_i.
\end{aligned}$$

Finally, the square of the cubic term in the Dirac operator is

$$\begin{aligned}
&\Big(\frac{1}{2}\sum_i X_i \cdot \widetilde{\mathrm{ad}}\, X_i\Big)^2 = \frac{1}{4}\sum_{i,j} X_i \cdot \widetilde{\mathrm{ad}}\, X_i \cdot X_j \cdot \widetilde{\mathrm{ad}}\, X_j \\
&= \frac{1}{4}\sum_{i,j} X_i \cdot X_j \cdot \widetilde{\mathrm{ad}}\, X_i \cdot \widetilde{\mathrm{ad}}\, X_j + \frac{1}{4}\sum_{i,j} X_i \cdot [\widetilde{\mathrm{ad}}\, X_i, X_j] \cdot \widetilde{\mathrm{ad}}\, X_j \\
&= -\frac{1}{4}\sum_i (\widetilde{\mathrm{ad}}\, X_i)^2 + \frac{1}{8}\sum_{i,j} X_i \cdot X_j \cdot \widetilde{\mathrm{ad}}[X_i, X_j] + \frac{1}{4}\sum_{i,j} X_i \cdot [X_i, X_j] \cdot \widetilde{\mathrm{ad}}\, X_j \\
&= -\frac{1}{4}\sum_i (\widetilde{\mathrm{ad}}\, X_i)^2 - \frac{1}{8}\sum_{i,j} X_i \cdot [X_j, X_i] \cdot \widetilde{\mathrm{ad}}\, X_j \\
&= -\frac{1}{4}\sum_i (\widetilde{\mathrm{ad}}\, X_i)^2 - \frac{1}{2}\sum_j (\widetilde{\mathrm{ad}}\, X_j)^2 = -\frac{3}{4}\sum_i (\widetilde{\mathrm{ad}}\, X_i)^2.
\end{aligned}$$

Putting all of these together, we obtain

$$\not{\partial}^2 = -\sum_i \Big(r(X_i)^2 \otimes 1 + r(X_i) \otimes \widetilde{\mathrm{ad}}\, X_i + 1 \otimes \frac{3}{4}(\widetilde{\mathrm{ad}}\, X_i)^2\Big). \tag{1.5}$$

To simplify this result, we introduce the Casimir operator, defined as the quadratic element

$$\Delta = -\sum_i X_i^* X_i \in U(\mathfrak{g}) \tag{1.6}$$

in the universal enveloping algebra of $\mathfrak{g}$. The Casimir operator Δ does not depend on the choice of basis for $\mathfrak{g}$, although it does depend on the inner product. Furthermore Δ commutes with all elements $X \in \mathfrak{g}$. Given any Lie

algebra homomorphism $\sigma : \mathfrak{g} \to \mathfrak{u}$, we write Δ_σ for $\sigma(\Delta)$. In terms of these Casimir operators, the computations above yield the following Weitzenböck formula for the square of the Dirac operator:

THEOREM 1. *The square of the Dirac operator on G is*

$$\not{\partial}^2 = \frac{1}{2}\Delta_r \otimes 1 + \frac{1}{2}\Delta_\varrho + 1 \otimes \frac{1}{4}\Delta_{\widetilde{\mathrm{ad}}}, \tag{1.7}$$

where $\varrho = r \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}$ is the right action of $\mathfrak{g}$ on spinors.

PROOF. Expanding the diagonal Casimir Δ_ϱ, we obtain

$$\Delta_\varrho = \Delta_r \otimes 1 - 2\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}\, X_i + 1 \otimes \Delta_{\widetilde{\mathrm{ad}}},$$

and the result follows immediately from (1.5) and (1.6). □

To understand the Dirac operator on G in terms of the representations of G, we must first examine the spin representation $\mathbb{S}$. The Lie algebra $\mathfrak{g}$ acts on $\mathbb{S}$ via the map $\widetilde{\mathrm{ad}} : \mathfrak{g} \to \mathfrak{spin}(\mathfrak{g})$. To compute the weights of this representation, we recall that the complexified Lie algebra decomposes as

$$\mathfrak{g}_\mathbb{C} = \mathfrak{t}_\mathbb{C} \oplus \bigoplus_{\alpha\in\Phi} \mathfrak{g}_\alpha,$$

where $\mathfrak{t}_\mathbb{C}$ is the Cartan subalgebra, $\Phi \subset \mathfrak{t}^*$ is the root system for $\mathfrak{g}$, and the $\mathfrak{g}_\alpha$ are the one dimensional root spaces satisfying $[H, X_\alpha] = i\alpha(H)X_\alpha$ for $H \in \mathfrak{t}_\mathbb{C}, X_\alpha \in \mathfrak{g}_\alpha$. The roots come in $\pm\alpha$ pairs, and we choose a positive root system Φ^+. To obtain a basis of $\mathfrak{g}_\mathbb{C}$ that respects this decomposition, we choose an orthonormal basis $\{H_1, \dots, H_k\}$ of $\mathfrak{t}$, and we pick generators $E_\alpha \in \mathfrak{g}_\alpha$ such that $\langle E_\alpha, E_{-\alpha}\rangle = \frac{1}{2}$. Given any $H \in \mathfrak{t}$, we have

$$\begin{aligned}
\widetilde{\mathrm{ad}}\, H &= \frac{1}{4}\Big(\sum_i H_i^* \cdot [H, H_i] + \sum_{\alpha\in\Phi} E_\alpha^* \cdot [H, E_\alpha]\Big) \\
&= \frac{1}{2}\sum_{\alpha\in\Phi} i\alpha(H) E_{-\alpha} \cdot E_\alpha \\
&= \frac{1}{2}\sum_{\alpha\in\Phi^+} i\alpha(H)\left(E_{-\alpha} \cdot E_\alpha - E_\alpha \cdot E_{-\alpha}\right) \\
&= i\rho(H) + \sum_{\alpha\in\Phi^+} i\alpha(H) E_{-\alpha} \cdot E_\alpha,
\end{aligned}$$

where $\rho = \frac{1}{2}\sum_{\alpha\in\Phi^+}\alpha$ is half the sum of the positive roots. Now, noting that $[\widetilde{\mathrm{ad}}\, H, E_\alpha] = (\mathrm{ad}\, H)E_\alpha = i\alpha(H)E_\alpha$, we see that applying E_α to a vector v shifts its weight by α. In particular, if v is a highest weight vector, then $E_\alpha v$ vanishes for all positive roots α, and so v must have weight ρ. It follows that

the spin representation $\mathbb{S}$ decomposes as a representation of $\mathfrak{g}$ into multiple copies of V_ρ, the irreducible representation with highest weight ρ.

Recall the *Peter-Weyl theorem*, which provides a decomposition of the space of L^2 functions on G into a sum of irreducible representations with respect to the action $l \otimes r$ of $G \times G$. Summing over all representations V_λ of G, or equivalently over all dominant weights λ, we have

$$L^2(G) \cong \widehat{\bigoplus}_\lambda V_\lambda \otimes V_\lambda^*,$$

where $\widehat{\bigoplus}$ denotes the Hilbert space direct sum. For L^2 spinors on G, we obtain a similar decomposition, now with respect to the action $l \otimes \varrho$ of $G \times \mathrm{G}$, given by

$$L^2(S) \cong \widehat{\bigoplus}_\lambda V_\lambda \otimes \big(V_\lambda^* \otimes \mathbb{S}\big).$$

Since the Dirac operator $\not{\partial}$ commutes with both the left action l and the right action ϱ on spinors, it takes a constant scalar value on each of the irreducible representations in this decomposition.

To compute the spectrum of the Dirac operator, we first look at the square of the Dirac operator given by (1.7). Like the Dirac operator, the Casimir operator Δ again commutes with the elements of $\mathfrak{g}$, so it also takes constant scalar values on the irreducible representations. If V_λ is the irreducible representation of $\mathfrak{g}$ with highest weight λ, then Δ takes on the value

$$\Delta_\lambda = \|\lambda + \rho\|^2 - \|\rho\|^2,$$

where ρ is half the sum of the positive roots. For example, on the spin representation $\mathbb{S}$, which is composed of multiple copies of V_ρ, we obtain $\Delta_{\widetilde{\mathrm{ad}}} = 3\|\rho\|^2$. In general, on the irreducible subrepresentations of the $V_\lambda^* \otimes (V_\lambda \otimes \mathbb{S})$ component (to simplify our computations we switch V_λ and V_λ^*, which makes no difference to the end result since we sum over all representations), we obtain

$$\not{\partial}^2 = \frac{1}{2}\|\lambda + \rho\|^2 + \frac{1}{2}\|\mu + \rho\|^2 - \frac{1}{4}\|\rho\|^2, \tag{1.8}$$

on the irreducible component $V_\mu \subset V_\lambda \otimes \mathbb{S}$ with highest weight μ.

Example. Consider the Dirac operator on the Lie group SU(2), whose underlying manifold is S^3. The Lie algebra $\mathfrak{su}(2)$ is generated by

$$X_1 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \qquad X_2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad X_3 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix},$$

which have the commutators $[X_1, X_2] = 2X_3$ (and cyclic permutations). For an inner product, we take $\langle X, Y\rangle = -\frac{1}{2}\operatorname{tr}(XY)$, which makes our basis orthonormal. The corresponding half-spin representation $\mathbb{S}^\pm$ each have complex dimension 2, and we can take the Clifford action of $\mathfrak{su}(2)$ on $\mathbb{S}^\pm$ to be simply

$c_\pm(X) = \pm X$. Furthermore, the spin lift of the adjoint representation is given by $\widetilde{\mathrm{ad}}(X) = X$ on both $\mathbb{S}^+$ and $\mathbb{S}^-$. The Dirac operators $\partial\!\!\!/_\pm$ on $\mathbb{S}^\pm$ are then

$$\partial\!\!\!/_\pm = \pm \sum_i X_i\, r(X_i) \mp \frac{3}{2},$$

where we compute the cubic term in the Dirac operator as follows:

$$\frac{1}{8}\sum_{i,j,k} X_i \cdot X_j \cdot [X_i, X_j] = \frac{3}{4} X_1 \cdot X_2 \cdot [X_1, X_2] = \pm\frac{3}{2} X_1 X_2 X_3 = \mp\frac{3}{2}.$$

Recalling the representation theory of SU(2), we have $\rho = 1$ and $\mathbb{S}^\pm \cong V_1$. Given any irreducible representation V_λ with $\lambda > 0$, the tensor product decomposes as $V_\lambda \otimes V_1 \cong V_{\lambda-1} \oplus V_{\lambda+1}$. Furthermore, we have $V_\lambda^* \cong V_\lambda$, and our inner product is scaled so that $\|\lambda\|^2 = \lambda^2$. So, on the irreducible subrepresentations of $V_\lambda \otimes (V_\lambda \otimes \mathbb{S})$, the expression (1.8) for the square of the Dirac operator becomes

$$\partial\!\!\!/^2 = \begin{cases} \lambda^2 + 3\lambda + \frac{9}{4} & \text{on } V_\lambda \otimes V_{\lambda+1} \text{ for } \lambda \geq 0, \\ \lambda^2 + \lambda + \frac{1}{4} & \text{on } V_\lambda \otimes V_{\lambda-1} \text{ for } \lambda \geq 1. \end{cases}$$

For the Dirac operator itself, we obtain both the positive and negative square roots, where the sign depends on which of the half-spin representations $\mathbb{S}^\pm$ we take. To determine the signs, we compute a few special cases. For example, on the representations $V_0 \otimes V_1^\pm$, the Dirac operator is just $\partial\!\!\!/_\pm = \mp\frac{3}{2}$. Also, noting that we can write the Dirac operator in the alternative form

$$\partial\!\!\!/_\pm = \pm \sum_i X_i \varrho(X_i) \pm \frac{3}{2},$$

we see that the Dirac operator takes on the value $\partial\!\!\!/_\pm = \pm\frac{3}{2}$ on the representations $V_1 \otimes V_0^\pm$. In general, we obtain the spectrum

$$\partial\!\!\!/ = \begin{cases} -\left(\lambda + \frac{3}{2}\right) & \text{on } V_\lambda \otimes V_{\lambda+1}^+ \text{ for } \lambda \geq 0, \\ +\left(\lambda + \frac{3}{2}\right) & \text{on } V_\lambda \otimes V_{\lambda+1}^- \text{ for } \lambda \geq 0, \\ +\left(\lambda + \frac{1}{2}\right) & \text{on } V_\lambda \otimes V_{\lambda-1}^+ \text{ for } \lambda \geq 1, \\ -\left(\lambda + \frac{1}{2}\right) & \text{on } V_\lambda \otimes V_{\lambda-1}^- \text{ for } \lambda \geq 1, \end{cases}$$

which completely describes the Dirac operator on SU(2).

1.3. The Dirac Operator on G/H

We now examine the Dirac operator on homogeneous spaces. As in Section 1.1, we let G be a compact, connected Lie group and H a closed connected subgroup, and we consider the manifold G/H. Letting $\mathfrak{p}$ be the orthogonal complement of $\mathfrak{h}$ in $\mathfrak{g}$ with respect to an Ad-invariant inner product, we recall

that the tangent bundle to G/H is the vector bundle $G \times_H \mathfrak{p}$ associated to the isotropy representation $\mathrm{Ad} : H \to \mathrm{SO}(\mathfrak{p})$. The corresponding frame bundle is the principal $\mathrm{SO}(\mathfrak{p})$-bundle $P \cong G \times_H \mathrm{SO}(\mathfrak{p})$.

The homogeneous space G/H is a spin manifold if and only if Ad can be lifted to a spin representation $\widetilde{\mathrm{Ad}} : H \to \mathrm{Spin}(\mathfrak{p})$. Given such a lift, the corresponding principal $\mathrm{Spin}(\mathfrak{p})$-bundle is then $Q \cong G \times_H \mathrm{Spin}(\mathfrak{p})$. Assuming $\dim \mathfrak{p} \geq 3$, since $\mathrm{Spin}(\mathfrak{p})$ is the universal cover of $\mathrm{SO}(\mathfrak{p})$, the map Ad admits a lift if and only if the induced map of fundamental groups

$$\mathrm{Ad}_* : \pi_1(H) \to \pi_1(\mathrm{SO}(\mathfrak{p})) \cong \mathbb{Z}_2$$

vanishes identically. Recalling that

$$H^1(H; \mathbb{Z}_2) \cong \mathrm{Hom}(\pi_1(H), \mathbb{Z}_2),$$

we see that Ad_* determines an element of $H^1(H; \mathbb{Z}_2)$, which is also the image of the generator of $H^1(\mathrm{SO}(\mathfrak{p}); \mathbb{Z}_2)$ via the pullback

$$\mathrm{Ad}^* : H^1(\mathrm{SO}(\mathfrak{p}); \mathbb{Z}_2) \to H^1(H; \mathbb{Z}_2).$$

In general, an oriented manifold M is spin if and only if $w_2 \in H^2(M; \mathbb{Z}_2)$ vanishes. To compare w_2 with our obstruction $\mathrm{Ad}_* \in H^1(H; \mathbb{Z}_2)$, consider the two fibrations

$$\begin{tikzcd} H \arrow[r] \arrow[d, "\mathrm{Ad}"] & G \arrow[r] \arrow[d] & G/H \arrow[d, "="] \\ \mathrm{SO}(\mathfrak{p}) \arrow[r] & P \arrow[r] & G/H \end{tikzcd}$$

Since the cohomology spectral sequence is functorial, we obtain the commutative diagram

$$\begin{tikzcd} H^1(\mathrm{SO}(\mathfrak{p}); \mathbb{Z}_2) \arrow[r, "d_2"] \arrow[d, "\mathrm{Ad}^*"] & H^2(G/H; \mathbb{Z}_2) \arrow[d, "="] \\ H^1(H; \mathbb{Z}_2) \arrow[r, "d_2"] & H^2(G/H; \mathbb{Z}_2) \end{tikzcd}$$

and it follows that $w_2 = d_2 \, \mathrm{Ad}_*$.

Example. If G is simply connected and T is a maximal torus in G, then G/T is a spin manifold. To show this, we first observe that the diagram

$$\begin{tikzcd} T \arrow[r, "\mathrm{Ad}"] \arrow[d] & \mathrm{SO}(\mathfrak{p}) \arrow[d, "i"] \\ G \arrow[r, "\mathrm{Ad}"] & \mathrm{SO}(\mathfrak{g}) \end{tikzcd}$$

commutes, where the vertical arrows are inclusions. Taking the induced maps on the fundamental groups, we obtain

$$\begin{array}{ccc} \pi_1(T) & \xrightarrow{\mathrm{Ad}_*} & \pi_1(\mathrm{SO}(\mathfrak{p})) \\ \downarrow & & \downarrow{\scriptstyle i_*} \\ 1 & \longrightarrow & \pi_1(\mathrm{SO}(\mathfrak{g})) \end{array}$$

Since this diagram commutes, and since $\ker i_* = 1$, the obstruction Ad_* must vanish and therefore G/T is a spin manifold.

If G/H is not a spin manifold, we can instead consider a Spin^c structure, replacing Spin in our discussion with the group

$$\mathrm{Spin}^c(n) = \mathrm{Spin}(n) \times \mathrm{U}(1) \,/\, \{\pm(1,1)\}.$$

Again taking $\dim \mathfrak{p} \geq 3$, we find that the isotropy representation $\mathrm{Ad} : H \to \mathrm{SO}(\mathfrak{p})$ can be lifted to a map $\widetilde{\mathrm{Ad}} : H \to \mathrm{Spin}^c(\mathfrak{p})$ if and only if the induced map on the fundamental groups

$$\mathrm{Ad}_* : \pi_1(H) \to \pi_1(\mathrm{SO}(\mathfrak{p})) \cong \mathbb{Z}_2$$

can be covered by a homomorphism

$$\widehat{\mathrm{Ad}}_* : \pi_1(H) \to \pi_1(\mathrm{Spin}^c(\mathfrak{p})) \cong \mathbb{Z}.$$

In other words, the manifold G/H is Spin^c if and only if there exists an element $\widehat{\mathrm{Ad}}_* \in H^1(H;\mathbb{Z})$ whose mod 2 reduction is Ad_*. In particular, if $\pi_1(H)$ has no 2-torsion, then G/H is a Spin^c manifold.

Example. Suppose G is simply connected, and let H be a maximal rank subgroup of G. By comparing the roots and unit lattices of G and H, we find that $\pi_1(H)$ must be torsion-free. Consequently, $\pi_1(H)$ has no 2-torsion and so G/H is therefore a Spin^c manifold. In fact, these homogeneous spaces are the complex flag manifolds, and they admit a canonical Spin^c structure coming from their complex structure.

Using the notation of Section 1.1, we write vector fields on G/H as H-equivariant maps $\xi : G \to \mathfrak{p}$. Recalling (1.1), the Levi-Civita connection on G/H is $\nabla_\xi = \partial_\xi + \frac{1}{2}\,\mathrm{ad}_{\mathfrak{p}}\,\xi$, where ∂_ξ is the directional derivative, and the map $\mathrm{ad}_{\mathfrak{p}} : \mathfrak{p} \to \mathfrak{so}(\mathfrak{p})$ is the projection of the adjoint representation onto $\mathfrak{p}$ given by $\mathrm{ad}_{\mathfrak{p}}\, X : Y \mapsto [X,Y]_{\mathfrak{p}}$ for $X, Y \in \mathfrak{p}$. Applying (1.2), the spin representation $\widetilde{\mathrm{ad}}_{\mathfrak{p}} : \mathfrak{p} \to \mathfrak{spin}(\mathfrak{p})$ lifting $\mathrm{ad}_{\mathfrak{p}}$ is

$$\widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X = \frac{1}{4} \sum_i X_i^* \cdot [X, X_i]_{\mathfrak{p}},$$

where the sum is taken over a basis X_i for $\mathfrak{p}$. The Levi-Civita connection on spinors then becomes $\nabla_\xi = \partial_\xi + \frac{1}{2}\,\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,\xi$.

In general, the Dirac operator at an individual point is given by $\not{\partial} = \sum_i c(e_i^*)\nabla_{e_i}$, where $\{e_i\}$ is a basis for the tangent space at that point. However, the Dirac operator itself does not depend on the choice of basis. On a homogeneous space G/H, given a basis $\{X_i\}$ for $\mathfrak{p}$, the left translates $(g, X_i) \in G \times_H \mathfrak{p}$ of this basis give a frame for the tangent space to G/H at the coset gH. If we choose a different representative for this coset, we obtain a distinct basis for the tangent space, but the Dirac operator remains unchanged. So, viewing the Dirac operator on G/H as an operator on the Lie group G, we may write it in terms of a frame of left-invariant vector fields on G. Note that these vector fields are not H-equivariant, so they do not descend to vector fields on G/H, but they do let us write the directional derivatives $\partial_{X_L} = r(X)$ in terms of the right action of $\mathfrak{g}$ on functions. The Dirac operator on G/H is thus the sum

$$\not{\partial} = \sum_i \Big(r(X_i) \otimes X_i^* + 1 \otimes \frac{1}{2} X_i^* \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i \Big), \tag{1.9}$$

taken over a basis for $\mathfrak{p}$. Algebraically, this Dirac operator can be viewed as an element of $U(\mathfrak{g}) \otimes \mathrm{Cl}(\mathfrak{p})$. Here, the cubic term of the Dirac operator can also be written in the form:

$$\frac{1}{2} X_i^* \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i = \frac{1}{8} \sum_{i,j} X_i^* \cdot X_j^* \cdot [X_i, X_j]_{\mathfrak{p}} = \frac{1}{8} \sum_{i,j,k} \langle [X_i, X_j], X_k \rangle X_i^* \cdot X_j^* \cdot X_k^*.$$

The following computations mirror those of the previous section. To avoid confusion, we denote elements of $\mathfrak{p}$ by X and Y, while we write Z for an element of $\mathfrak{h}$. In particular, we let X_i be a basis for $\mathfrak{p}$ and use indices i, j, k, while we write Z_a for a basis of $\mathfrak{h}$. In this notation, the spin lift of the isotropy representation $\mathrm{ad} : \mathfrak{h} \to \mathfrak{so}(\mathfrak{p})$ is the map

$$\widetilde{\mathrm{ad}} : Z \to \frac{1}{4} \sum_i X_i^* \cdot [Z, X_i], \tag{1.10}$$

where $[Z, X_i]$ is in $\mathfrak{p}$ because the inner product is Ad-invariant. Working in the Clifford algebra $\mathrm{Cl}(\mathfrak{p})$, we then have the commutators

$$\begin{aligned} [\widetilde{\mathrm{ad}}_{\mathfrak{p}} X, Y] &= [X, Y]_{\mathfrak{p}}, \\ [\widetilde{\mathrm{ad}}\, Z, X] &= [Z, X], \\ [\widetilde{\mathrm{ad}}\, Z, \widetilde{\mathrm{ad}}_{\mathfrak{p}} X] &= \widetilde{\mathrm{ad}}_{\mathfrak{p}} [Z, X]. \end{aligned}$$

The third commutator follows from the first two and the Jacobi identity:

$$\begin{aligned} [[\widetilde{\mathrm{ad}}\, Z, \widetilde{\mathrm{ad}}_{\mathfrak{p}} X], Y] &= [\widetilde{\mathrm{ad}}\, Z, [\widetilde{\mathrm{ad}}_{\mathfrak{p}} X, Y]] - [\widetilde{\mathrm{ad}}_{\mathfrak{p}} X, [\widetilde{\mathrm{ad}}\, Z, Y]] \\ &= [Z, [X, Y]_{\mathfrak{p}}] - [X, [Z, Y]]_{\mathfrak{p}} \\ &= [Z, [X, Y]]_{\mathfrak{p}} - [X, [Z, Y]]_{\mathfrak{p}} \\ &= [[Z, X], Y]_{\mathfrak{p}}, \end{aligned}$$

recalling that the map $\mathrm{ad} : \mathfrak{spin}(\mathfrak{p}) \to \mathfrak{so}(\mathfrak{p})$ is an isomorphism. We also have an immediate generalization of Lemma 2 to the homogeneous case, which provides the two identities

$$\sum_i b\big([X_i, Z], X_i^*\big) = \sum_i b\big(X_i, [Z, X_i^*]\big), \tag{1.11}$$

$$\sum_i b\big([X_i, Y]_{\mathfrak{p}}, X_i^*\big) = \sum_i b\big(X_i, [Y, X_i^*]_{\mathfrak{p}}\big), \tag{1.12}$$

for any bilinear map $b : \mathfrak{p} \times \mathfrak{p} \to V$ taking values in a vector space V. To prove these identities, we compose b with the orthogonal projection $\mathfrak{g} \times \mathfrak{g} \to \mathfrak{p} \times \mathfrak{p}$ and apply Lemma 2 as before. Alternatively, if the bilinear map is of the form $b : \mathfrak{g} \times \mathfrak{p} \to V$, the identity becomes

$$\sum_i b([X_i, Y], X_i^*) = \sum_i b(X_i, [Y, X_i^*]_p) + \sum_a b(Z_a, [Y, Z_a^*]). \tag{1.13}$$

Recall that we are viewing spinors on G/H as maps $s : G \to \mathbb{S}$ on G satisfying the H-equivariance condition $s(gh) = (\widetilde{\mathrm{Ad}}\, h^{-1})s(g)$. Equivalently, we may consider maps that are invariant under the right H-action $(\varrho(h)s)(g) = (\widetilde{\mathrm{Ad}}\, h)s(gh)$ on spinors. At the Lie algebra level, this induces the $\mathfrak{h}$-action $\varrho(Z) = r(Z) \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\, Z$. In order to confirm that our Dirac operator is well-defined as an operator on G/H, we must verify that it commutes with this right action ϱ.

Lemma 4. *The Dirac operator $\partial\!\!\!/$ on G/H commutes with the right action*

$$\varrho(Z) = r(Z) \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\, Z \in U(\mathfrak{g}) \otimes \mathrm{Cl}(\mathfrak{p})$$

for all $Z \in \mathfrak{h}$.

Proof. As in the proof of Lemma 3, we compute first

$$\begin{aligned}\Big[\varrho(Z), \sum_i r(X_i) \otimes X_i^*\Big] &= \sum_i \big([r(Z), r(X_i)] \otimes X_i^* + r(X_i) \otimes [\widetilde{\mathrm{ad}}\, Z, X_i^*]\big)\\ &= \sum_i \big(r([Z, X_i]) \otimes X_i^* + r(X_i) \otimes [Z, X_i^*]\big)\\ &= 0,\end{aligned}$$

and then for the cubic term of the Dirac operator we have

$$\begin{aligned}\Big[\widetilde{\mathrm{ad}}\, Z, \sum_i X_i^* \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i\Big] &= \sum_i \big([\widetilde{\mathrm{ad}}\, Z, X_i^*] \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i + X_i^* \cdot [\widetilde{\mathrm{ad}}\, Z, \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i]\big)\\ &= \sum_i \big([Z, X_i^*] \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i + X_i^* \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}[Z, X_i]\big)\\ &= 0.\end{aligned}$$

Combining these two commutators, we obtain $[\varrho(Z), \partial\!\!\!/] \equiv 0$. $\square$

Once again, we compute the square of the Dirac operator. We work with orthonormal bases $\{X_i\}$ for $\mathfrak{p}$ and $\{Z_a\}$ for $\mathfrak{h}$. We also write $A_i = \frac{1}{2}\,\widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i$ for the connection 1-form, so our Dirac operator becomes $\not{\partial} = r(X_i) \otimes X_i + 1 \otimes X_i \cdot A_i$. Taking the first portion of the Dirac operator, its square is

$$\begin{aligned}\Big(\sum\nolimits_i r(X_i) \otimes X_i\Big)^2 &= \sum\nolimits_{i,j} r(X_i) r(X_j) \otimes X_i \cdot X_j \\ &= \sum\nolimits_i r(X_i)^2 \otimes X_i^2 + \frac{1}{2}\sum\nolimits_{i,j} [r(X_i), r(X_j)] \otimes X_i \cdot X_j \\ &= -\sum\nolimits_i r(X_i)^2 \otimes 1 + \frac{1}{2}\sum\nolimits_{i,j} r([X_i, X_j]) \otimes X_i \cdot X_j.\end{aligned}$$

Applying (1.13) to the second term, we obtain

$$\begin{aligned}\frac{1}{2}\sum_{i,j} r([X_i, X_j]) \otimes X_i \cdot X_j &= \frac{1}{2}\sum_{i,j} r(X_i) \otimes [X_j, X_i]_{\mathfrak{p}} \cdot X_j \\ &\qquad + \frac{1}{2}\sum_{a,j} r(Z_a) \otimes [X_j, Z_a]_{\mathfrak{p}} \cdot X_j \\ &= 2\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i + 2\sum_a r(Z_a) \otimes \widetilde{\mathrm{ad}}\, Z_a.\end{aligned}$$

The cross terms in the square of the Dirac operator are

$$\begin{aligned}\Big\{\sum_i r(X_i) \otimes X_i, \sum_j 1 \otimes X_j \cdot A_j\Big\} &= \sum_{i,j} r(X_i) \otimes \{X_i, X_j \cdot A_j\} \\ &= \sum_{i,j} r(X_i) \otimes \big(\{X_i, X_j\} \cdot A_j - X_j \cdot [X_i, A_j]\big) \\ &= -2\sum_i r(X_i) \otimes A_i - \sum_{i,j} r(X_i) \otimes X_j \cdot [X_i, A_j].\end{aligned}$$

The square of the cubic term in the Dirac operator is

$$\begin{aligned}\Big(\sum_i X_i \cdot A_i\Big)^2 &= \sum_{i,j} X_i \cdot A_i \cdot X_j \cdot A_j \\ &= \sum_{i,j} X_i \cdot X_j \cdot A_i \cdot A_j + \sum_{i,j} X_i \cdot [A_i, X_j] \cdot A_j \\ &= -\sum_i A_i^2 + \frac{1}{2}\sum_{i,j} X_i \cdot X_j \cdot [A_i, A_j] + \sum_{i,j} X_i \cdot [A_i, X_j] \cdot A_j.\end{aligned}$$

Noting that $X_j \cdot [X_i, A_j] = 4A_i$, these three terms combine to give

$$\begin{aligned}\not{\partial}^2 = &-\sum_i r(X_i)^2 \otimes 1 + 2\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i + 2\sum_a r(Z_a) \otimes \widetilde{\mathrm{ad}}\, Z_a \\ &- 6\sum_i r(X_i) \otimes A_i - 5\sum_i 1 \otimes A_i^2 + 1 \otimes c(F_A),\end{aligned}$$

where $c(F_A) = \frac{1}{2}\sum_{i,j} X_i \cdot X_j \cdot [A_i, A_j]$ is the element of the Clifford algebra corresponding to the curvature F_A of our connection A via the identification $\Lambda^2(\mathfrak{p}) \cong \mathfrak{spin}(\mathfrak{p}) \subset \mathrm{Cl}(\mathfrak{p})$. Expanding this curvature term, we have

$$\begin{aligned}c(F_A) &= \frac{1}{8}\sum_{i,j} X_i \cdot X_j \cdot [\widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i, \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_j] \\ &= \frac{1}{32}\sum_{i,j,k} X_i \cdot X_j \cdot X_k \cdot \big([X_i, [X_j, X_k]_{\mathfrak{p}}]_{\mathfrak{p}} - [X_j, [X_i, X_k]_{\mathfrak{p}}]_{\mathfrak{p}}\big) \\ &= \frac{1}{16}\sum_{i,j,k} X_j \cdot X_k \cdot X_i \cdot [X_i, [X_j, X_k]_{\mathfrak{p}}]_{\mathfrak{p}} - \frac{3}{16}\sum_{i,j} X_j \cdot [X_i, [X_i, X_j]_{\mathfrak{p}}]_{\mathfrak{p}} \\ &= \frac{1}{16}\sum_{i,j,k} X_j \cdot [X_k, X_j]_{\mathfrak{p}} \cdot X_i \cdot [X_i, X_k]_{\mathfrak{p}} - \frac{3}{16}\sum_{i,j} [X_j, X_i]_{\mathfrak{p}} \cdot [X_i, X_j]_{\mathfrak{p}} \\ &= -\sum_i (\widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i)^2 - \frac{3}{16}\sum_{i,j} \langle [X_i, X_j]_{\mathfrak{p}}, [X_i, X_j]_{\mathfrak{p}} \rangle.\end{aligned}$$

The scalar term in this expression simplifies further if we apply (1.12) (or alternatively if we use the ad-invariance of the metric), giving us

$$\sum_{i,j} \langle [X_i, X_j]_{\mathfrak{p}}, [X_i, X_j]_{\mathfrak{p}} \rangle = \sum_{i,j} \langle [X_i, [X_j, X_i]_{\mathfrak{p}}]_{\mathfrak{p}}, X_j \rangle = -\operatorname{tr}_{\mathfrak{p}} \sum_i (\mathrm{ad}_{\mathfrak{p}} X_i)^2.$$

Combining all of the computations above, we find that the square of the Dirac operator on G/H is

$$\begin{aligned}\not{\partial}^2 = &-\sum_i r(X_i)^2 \otimes 1 - \sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i + 2\sum_a r(Z_a) \otimes \widetilde{\mathrm{ad}}\, Z_a \\ &- 1 \otimes \Big(\frac{9}{4}\sum_i (\widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i)^2 - \frac{3}{16}\operatorname{tr}_{\mathfrak{p}} \sum_i (\mathrm{ad}_{\mathfrak{p}} X_i)^2\Big).\end{aligned}$$

This expression for the square of the Dirac operator is rather unwieldy. However, if we change the Dirac operator slightly by using a different connection, we can choose a new Dirac operator with a much simpler and useful expression for its square. In particular, we can eliminate the $\sum_i r(X_i) \otimes \widetilde{\mathrm{ad}}_{\mathfrak{p}} X_i$

term by taking the Dirac operator corresponding to the spin connection $A_i' = \frac{2}{3}A_i = \frac{1}{3}\,\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,X_i$. This gives the Dirac operator

$$\Box = \sum_i r(X_i)\otimes X_i^* + 1\otimes\frac{1}{12}\sum_{i,j} X_i^*\cdot X_j^*\cdot[X_i,X_j], \tag{1.14}$$

which is examined by Kostant in [**4**]. The square of Kostant's Dirac operator is then

$$\begin{aligned}\Box^2 = &-\sum_i r(X_i)^2\otimes 1 + 2\sum_a r(Z_a)\otimes\widetilde{\mathrm{ad}}\,Z_a\\ &-1\otimes\Big(\sum_i(\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,X_i)^2 - \frac{1}{12}\,\mathrm{tr}_{\mathfrak{p}}\sum_i(\mathrm{ad}_{\mathfrak{p}}\,X_i)^2\Big)\end{aligned}$$

Writing this in terms of the Casimir operators $\Delta_r^{\mathfrak{g}}$ for the right action of $\mathfrak{g}$ on functions and $\Delta_\varrho^{\mathfrak{h}}$ for the diagonal right action $\varrho = r\otimes 1 + 1\otimes\widetilde{\mathrm{ad}}$ of $\mathfrak{h}$ on spinors, we obtain

$$\Box^2 = \Delta_r^{\mathfrak{g}}\otimes 1 - \Delta_\varrho^{\mathfrak{h}} - 1\otimes\Big(\sum_a(\widetilde{\mathrm{ad}}\,Z_a)^2 + \sum_i(\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,X_i)^2 - \frac{1}{12}\,\mathrm{tr}_{\mathfrak{p}}\sum_i(\mathrm{ad}_{\mathfrak{p}}\,X_i)^2\Big).$$

We can simplify this expression further by expanding the term $\sum_a(\widetilde{\mathrm{ad}}\,Z_a)^2 + \sum_i(\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,X_i)^2$. This is a sum of squares of quadratic elements in the Clifford algebra. Now, if ω is a quadratic element, we have $a(\omega) = -\omega$, where a is the fundamental anti-automorphism reversing the order of Clifford multiplication. On the other hand, a leaves the degree 0 and 4 elements of the Clifford algebra fixed. We then have $a(\omega^2) = (a(\omega))^2 = \omega^2$, so we see that ω^2 must have components of degree 0 and 4, but no quadratic component. Using this result as well as the Jacobi identity, we compute

$$\begin{aligned}\sum\nolimits_a(\widetilde{\mathrm{ad}}\,Z_a)^2 + \sum\nolimits_i(\widetilde{\mathrm{ad}}_{\mathfrak{p}}\,X_i)^2 &= \frac{1}{16}\sum\nolimits_{a,j,k} X_j\cdot[Z_a,X_j]_{\mathfrak{p}}\cdot X_k\cdot[Z_a,X_k]_{\mathfrak{p}}\\ &\quad+\frac{1}{16}\sum\nolimits_{i,j,k} X_j\cdot[X_i,X_j]_{\mathfrak{p}}\cdot X_k\cdot[X_i,X_k]_{\mathfrak{p}}\\ &= \frac{1}{16}\sum\nolimits_{i,j,k} X_j\cdot X_i\cdot X_k\cdot[[X_j,X_i],X_k]_{\mathfrak{p}}\\ &= -\frac{1}{8}\sum\nolimits_{i,j} X_j\cdot[[X_j,X_i],X_i]_{\mathfrak{p}}\\ &= \frac{1}{8}\sum\nolimits_{i,j}\langle X_j,[[X_j,X_i],X_i]\rangle = \frac{1}{8}\,\mathrm{tr}_{\mathfrak{p}}\sum\nolimits_i(\mathrm{ad}\,X_i)^2\\ &= \frac{1}{8}\,\mathrm{tr}_{\mathfrak{p}}\sum\nolimits_i\big((\mathrm{ad}_{\mathfrak{p}}\,X_i)^2 + (\mathrm{ad}\,X_i)(\mathrm{ad}_{\mathfrak{h}}\,X_i)\big).\end{aligned}$$

Now, we note that since the inner product on $\mathfrak{g}$ is Ad-invariant, the following three trace expressions are all equal to each other:

$$\begin{aligned}
\operatorname{tr}_{\mathfrak{p}} \sum\nolimits_a (\operatorname{ad} Z_a)^2 &= \sum\nolimits_{a,i,j} \langle [Z_a, X_j], X_i \rangle \langle [Z_a, X_i], X_j \rangle, \\
\operatorname{tr}_{\mathfrak{h}} \sum\nolimits_i (\operatorname{ad} X_i)^2 &= \sum\nolimits_{a,i,j} \langle [X_i, X_j], Z_a \rangle \langle [X_i, Z_a], X_j \rangle, \\
\operatorname{tr}_{\mathfrak{p}} \sum\nolimits_i (\operatorname{ad} X_i)(\operatorname{ad}_h X_i) &= \sum\nolimits_{a,i,j} \langle [X_i, Z_a], X_j \rangle \langle [X_i, X_j], Z_a \rangle.
\end{aligned}$$

Combining these three expressions, we find that

$$\begin{aligned}
\operatorname{tr}_{\mathfrak{g}} \Delta_{\mathrm{ad}}^{\mathfrak{g}} - \operatorname{tr}_{\mathfrak{h}} \Delta_{\mathrm{ad}}^{\mathfrak{h}} &= -\operatorname{tr}_{\mathfrak{p}} \sum\nolimits_i (\operatorname{ad}_{\mathfrak{p}} X_i)^2 - \operatorname{tr}_{\mathfrak{p}} \sum\nolimits_a (\operatorname{ad} Z_a)^2 \\
&\quad - \operatorname{tr}_{\mathfrak{h}} \sum\nolimits_i (\operatorname{ad} X_i)^2 - \operatorname{tr}_{\mathfrak{p}} \sum\nolimits_i (\operatorname{ad} X_i)(\operatorname{ad} X_i)_{\mathfrak{h}} \\
&= -\operatorname{tr}_{\mathfrak{p}} \sum\nolimits_i (\operatorname{ad}_{\mathfrak{p}} X_i)^2 - 3 \operatorname{tr}_{\mathfrak{p}} \sum\nolimits_a (\operatorname{ad} Z_a)^2.
\end{aligned}$$

Putting all of these computations together, we obtain

$$\begin{aligned}
\Box^2 &= \Delta_r^{\mathfrak{g}} \otimes 1 - \Delta_\varrho^{\mathfrak{h}} + \frac{1}{24} \big(\operatorname{tr}_{\mathfrak{g}} \Delta_{\mathrm{ad}}^{\mathfrak{g}} - \operatorname{tr}_{\mathfrak{h}} \Delta_{\mathrm{ad}}^{\mathfrak{h}} \big) 1 \otimes 1 \\
&= \Delta_r^{\mathfrak{g}} \otimes 1 - \Delta_\varrho^{\mathfrak{h}} + \big(\|\rho_G\|^2 - \|\rho_H\|^2 \big)\, 1 \otimes 1.
\end{aligned}$$

To obtain this final expression, we use the identity $\frac{1}{24} \operatorname{tr} \Delta_{\mathrm{ad}} = \|\rho\|^2$. One way to prove this identity is to note first that, via a computation similar to the one on the previous page, we have $\Delta_{\widetilde{\mathrm{ad}}} = \frac{1}{8} \operatorname{tr} \Delta_{\mathrm{ad}}$. Then we recall from Section 1.2 that since the spin representation for a Lie algebra has highest weight ρ, its Casimir operator $\Delta_{\widetilde{\mathrm{ad}}}$ takes on the value $3\|\rho\|^2$.

We can now compute the spectrum of Kostant's Dirac operator. Regarding spinors as maps from G to $\mathbb{S}$, we recall that the H-equivariance condition is equivalent to saying that our maps are H-invariant with regard to the diagonal action ϱ. In particular, this means that the Casimir operator $\Delta_\varrho^{\mathfrak{h}}$ vanishes, so the square of Kostant's Dirac operator when applied to spinors on G/H is simply

$$\Box^2 = \Delta_r^{\mathfrak{g}} + \|\rho_G\|^2 - \|\rho_H\|^2.$$

Furthermore, on the Peter-Weyl component $V_\lambda^* \otimes (V_\lambda \otimes \mathbb{S})^{\varrho(\mathfrak{h})}$, we have

$$\Box^2 = \|\lambda + \rho_G\|^2 - \|\rho_H\|^2.$$

The spectrum of the Dirac operator itself is then obtained by taking the positive and negative square roots these values.

1.4. The Index of the Dirac Operator

Given any H-action ρ on a vector space V, we consider the associated vector bundle $G \times_H V$ over the homogeneous space G/H. The sections of this bundle are the equivariant V-valued functions on G satisfying $s(gh) = \rho(h)^{-1} s(g)$ for

all $h \in H$. Applying the Peter-Weyl theorem, we can decompose the space of L^2 maps from G to V into a sum over the irreducible representations V_λ of G,

$$L^2(G) \otimes V \cong \widehat{\bigoplus}_\lambda V_\lambda \otimes V_\lambda^* \otimes V,$$

with respect to the $G \times G \times H$ action $l \otimes r \otimes \rho$. The H-equivariance condition is equivalent to requiring that the functions be invariant under the diagonal H-action $h \mapsto r(h) \otimes \rho(h)$. Restricting the Peter-Weyl decomposition above to the H-invariant subspace, we obtain

$$\begin{aligned} L^2(G \times_H V) &\cong \widehat{\bigoplus}_\lambda V_\lambda \otimes (V_\lambda^* \otimes V)^H \\ &\cong \widehat{\bigoplus}_\lambda V_\lambda \otimes \mathrm{Hom}_H(V_\lambda, V). \end{aligned}$$

The Lie group G acts on the space of sections by $l(g)$, the left action of G on functions, which is preserved by this construction. The space $L^2(G \times_H V)$ is thus an infinite dimensional representation of G.

To formalize the discussion in the previous paragraph, we introduce the representation ring $R(G)$ of isomorphism classes of virtual G-representations. The elements of the representation ring are the finite linear combinations $\sum_\lambda a_\lambda V_\lambda$ with $a_\lambda \in \mathbb{Z}$, generated by the irreducible representations V_λ. The intertwining number of two representations gives us an inner product

$$\langle V, W \rangle_G = \dim_{\mathbb{C}} \mathrm{Hom}_G(V, W)$$

on $R(G)$, with respect to which the irreducible representations form an orthonormal basis. Taking the formal completion of $R(G)$, we define $\hat{R}(G)$ to be the group of possibly infinite formal sums $\sum_\lambda a_\lambda V_\lambda$. The intertwining number then extends to a pairing $R(G) \times \hat{R}(G) \to \mathbb{Z}$.

If H is a subgroup of G, every representation of G automatically restricts to a representation of H. This gives us a pullback map $i^* : R(G) \to R(H)$, corresponding to the inclusion $i : H \hookrightarrow G$. The map $V \mapsto L^2(G \times_H V)$ discussed above assigns to each H-representation an induced infinite dimensional G-representation. Expressed in terms of our representation ring notation, this induction map becomes the homomorphism $i_* : R(H) \to \hat{R}(G)$ given by

$$i_* V = \sum_\lambda \langle i^* V_\lambda, V \rangle_H V_\lambda,$$

which is formally adjoint to the pullback i^*.

A *homogeneous differential operator* on G/H is a linear differential operator $D : \Gamma(E) \to \Gamma(F)$ between two homogeneous vector bundles E and F that commutes with the left action of G on sections. If the operator is elliptic, then its kernel and cokernel are both finite dimensional representations of G, and

thus its G-index is a virtual representation in $R(G)$. In this case, the index takes a particularly elegant form.

THEOREM 2 (Bott). *If $D : \Gamma(G \times_H V) \to \Gamma(G \times_H W)$ is an elliptic homogeneous differential operator, then the G-equivariant index of D is given by*

$$\operatorname{Index}_G D = i_*(V - W),$$

where $i_*(V - W)$ is a finite element in $R(G)$.

In particular, note that the index of a homogeneous differential operator depends only on the vector bundles involved and not on the operator itself! In [**2**], Bott proves this using the Peter-Weyl theorem to decompose the original operator into an infinite family of finite dimensional operators indexed by the irreducible representations of G. Given a linear transformation $T : A \to B$ between finite dimensional vector spaces, the alternating sum of the dimensions of the terms in the exact sequence

$$0 \longrightarrow \operatorname{Ker} T \longrightarrow A \xrightarrow{T} B \longrightarrow \operatorname{Coker} T \longrightarrow 0$$

vanishes, and we obtain

$$\operatorname{Index} T = \dim \operatorname{Ker} T - \dim \operatorname{Coker} T = \dim A - \dim B.$$

Adding the equivariant indices for all of the finite dimensional components of D, we see that the index of D must be

$$\operatorname{Index}_G D = i_* V - i_* W = i_*(V - W),$$

the difference between the domain and range of D.

Let $\mathbb{S} = \mathbb{S}^+ \oplus \mathbb{S}^-$ be the spin representation of $\operatorname{Spin}(\mathfrak{p})$. If G/H is a spin manifold, then we can compose this spin representation with our lifted adjoint map $\widetilde{\operatorname{Ad}} : H \to \operatorname{Spin}(\mathfrak{p})$ to realize $\mathbb{S}$ as an representation of H. On the other hand, if G/H is a Spin^c manifold, then the adjoint representation is covered by a map

$$\widetilde{\operatorname{Ad}} : H \to \operatorname{Spin}^c(\mathfrak{p}) = \operatorname{Spin}(\mathfrak{p}) \times \mathrm{U}(1) \,/\, \{\pm(1,1)\}.$$

The $\operatorname{Spin}(\mathfrak{p})$ component acts on $\mathbb{S}$ via the standard spin representation while the $\mathrm{U}(1)$ component acts on $\mathbb{S}$ by complex multiplication. The diagonal element $(-1,-1)$ acts trivially, so again we obtain a unitary Spin^c-action and in turn a unitary H-action on $\mathbb{S}$. At the Lie algebra level, the adjoint action has the spin lift $\widetilde{\mathrm{ad}} : \mathfrak{h} \to \mathfrak{spin}(\mathfrak{p})$ given by (1.10) regardless of whether or not the manifold is spin. In the Spin^c case, we also have a representation $\mathrm{ad}_0\, \mathfrak{h} \to \mathfrak{u}(1)$ induced by the $\mathrm{U}(1)$ component of $\widetilde{\operatorname{Ad}}$. We denote this one dimensional $\mathfrak{h}$-representation by $\mathbb{C}_\nu$, where ν is its weight. The representations $\mathbb{S} \otimes \mathbb{C}_\nu$ and $\mathbb{C}^2_\nu$ of the Lie algebra $\mathfrak{h}$ both induce representations of the Lie group

H. However, if G/H is not spin, then neither $\mathbb{S}$ nor $\mathbb{C}_\nu$ alone corresponds to an H-representation.

We now consider the case where G is semi-simple and simply connected, and H is a reductive subgroup of maximal rank. We write ρ_H and ρ_G for half the sum of the positive roots of H and G respectively. To specify our sign convention for the half-spin representations, we note that the $\mathfrak{h}$-representation $\mathbb{S}$ has highest weight $\rho_G - \rho_H$, and that the corresponding weight space is one dimensional. We take $\mathbb{S}^+$ to be the half-spin representation containing that highest weight space. The operator that we are interested in is the twisted Dirac operator

$$\partial\!\!\!/^* \otimes U : \Gamma\left(G \times_H \left((\mathbb{S}^+)^* \otimes U\right)\right) \to \Gamma\left(G \times_H \left((\mathbb{S}^-)^* \otimes U\right)\right), \tag{1.15}$$

where U is a representation of the Lie algebra $\mathfrak{h}$ such that $\mathbb{S} \otimes U$ gives a representation of the Lie group H. This operator is given by (1.9) with an additonal term $\sum_i X_i^* A_i$ coming from the connection 1-form A_i on the U component. Note that this choice of connection does not change the index of the Dirac operator. Also, we use the adjoints $(\mathbb{S}^\pm)^*$ of the half-spin representations for reasons that will become clear in the proof of Theorem 3 below. Note that $(\mathbb{S}^\pm)^* = \mathbb{S}^\pm$ if $\frac{1}{2}\dim G/H$ is even, and that $(\mathbb{S}^\pm)^* = \mathbb{S}^\mp$ if it is odd.

The following theorem gives an expression for the index of this twisted Dirac operator in the case where G/H is spin. In particular, if we consider G/T, where T is a maximal torus, then this theorem becomes a version of the Borel-Weil-Bott theorem expressed in terms of spinors and the Dirac operator, instead of in its customary form involving holomorphic sections and Dolbeault cohomology.

THEOREM 3. *Let G/H be a spin manifold, and consider the Dirac operator $\partial\!\!\!/^* \otimes U_\mu$ twisted by an irreducible representation U_μ of H with highest weight μ. The G-equivariant index of this Dirac operator is the virtual G-representation*

$$\mathrm{Index}_G\ \partial\!\!\!/^* \otimes U_\mu = (-1)^w V_{w(\mu+\rho_H)-\rho_G} \tag{1.16}$$

if there exists an element $w \in W_G$ in the Weyl group of G such that the weight $w(\mu+\rho_H)-\rho_G$ is dominant for G. If no such w exists, then $\mathrm{Index}_G\ \partial\!\!\!/^* \otimes U_\mu = 0$.

PROOF. Adopting the notation of [**2**], we write

$$\Omega = \chi(\mathbb{S}^+ - \mathbb{S}^-) = \prod\nolimits_{\alpha \in \Phi^+_{G/H}} \left(e^{\alpha/2} - e^{-\alpha/2}\right),$$

where χ denotes the character of a representation, and $\Phi^+_{G/H}$ is the system of positive roots of G that are not roots of H. We also define the alternating elements $A_\lambda = \sum_{w \in W} (-1)^w e^{w(\lambda)}$, and we recall the Weyl character formula

$$\chi(V_\lambda) A_\rho = A_{\lambda+\rho}.$$

Recalling the identity

$$A_\rho = \sum_{w\in W}(-1)^w e^{w(\rho)} = \prod_{\alpha\in\Phi^+}\left(e^{\alpha/2}-e^{-\alpha/2}\right),$$

we observe that $A^G_{\rho_G} = A^H_{\rho_H}\Omega$. In [**2**], Bott proves the identity

$$\sum_{w\in W_G/W_H}(-1)^w w\left(A^H_{\rho_H}\chi(U_\mu)\right) = A^G_{\rho_G}\chi\big(i_*(\Omega^* U_\mu)\big). \tag{1.17}$$

Expanding the left side using the Weyl character formula on H, we have

$$\sum_{w\in W_G/W_H}(-1)^w w\left(A^H_{\rho_H}\chi(U_\mu)\right) = \sum_{w\in W_G/W_H}(-1)^w w(A^H_{\mu+\rho_H}) = A^G_{\mu+\rho_H}$$

At this point, there are two possibilities. If $\mu+\rho_H$ lies on the border of a Weyl chamber, then the alternating sum $A^G_{\mu+\rho_H}$ vanishes. Otherwise, $\mu+\rho_H$ must lie in the interior of some Weyl chamber. Since the Weyl group W_G acts simply transitively on the Weyl chambers, there exists a unique element $w\in W_G$ mapping $\mu+\rho_H$ into the positive Weyl chamber for G, and we have $A^G_{\mu+\rho_H} = (-1)^w A^G_{w(\mu+\rho_H)}$. This gives us the identity

$$\chi\big(i_*(\Omega^* U_\mu)\big)A^G_{\rho_G} = (-1)^w A^G_{w(\mu+\rho_H)}.$$

Applying the Weyl character formula for G, we find that

$$i_*(\Omega^* U_\mu) = (-1)^w V_{w(\mu+\rho_H)-\rho_G}.$$

Finally, by Theorem 2, the index of our Dirac operator is

$$\mathrm{Index}_G\, \partial\!\!\!/^* \otimes U_\mu = i_*\left((\mathbb{S}^+)^* U_\mu - (\mathbb{S}^-)^* U_\mu\right) = i_*\left(\Omega^* U\mu\right),$$

which gives us the desired result. □

In general, if G/H is not a spin manifold, then in order to obtain a similar result we must instead consider the operator

$$\partial\!\!\!/^* \otimes U_\mu : \left(L^2(G)\otimes(\mathbb{S}^+)^*\otimes U_\mu\right)^{\mathfrak{h}} \to \left(L^2(G)\otimes(\mathbb{S}^-)^*\otimes U_\mu\right)^{\mathfrak{h}},$$

viewed as an operator on G, restricted to the space of $(\mathbb{S}\otimes U_\mu)$-valued functions on G that are invariant under the diagonal $\mathfrak{h}$-action $\varrho(Z) = r(Z) + \widetilde{\mathrm{ad}}\, Z + \sigma(Z)$, where σ is the $\mathfrak{h}$-action on U_μ. Note that if $\mathbb{S}\otimes U_\mu$ is induced by a representation of the Lie group H, then this operator descends to a well-defined operator on G/H as before. In general, the G-equivariant index of this operator $\partial\!\!\!/^* \otimes U_\mu$ is once again given by (1.16). To prove this, we note that Bott's identity (1.17) continues to hold for the induction map $i_* : R(\mathfrak{h}) \to \hat{R}(\mathfrak{g})$ using the representation rings for the Lie algebras instead of the Lie groups. Working in the Lie algebra context, we no longer need concern ourselves with the topological obstructions occuring in the global Lie group picture. The rest of the proof of Theorem 3 continues unchanged.

In [**3**], Gross, Kostant, Ramond, and Sternberg note a generalization of the Weyl character formula for a maximal rank subgroup H of G. Let $C \subset W_G$ be the subset of elements that map the positive Weyl chamber for G into the positive Weyl chamber for H. The number of elements in C is the index of W_H in W_G, which is in turn the Euler characteristic $\chi(G/H)$ of the homogeneous space G/H. Working in $R(H)$, if V_λ is the representation of G with highest weight λ, we have

$$V_\lambda \otimes \mathbb{S}^+ - V_\lambda \otimes \mathbb{S}^- = \sum_{c \in C} (-1)^c \, U_{c \bullet \lambda}, \tag{1.18}$$

where $c \bullet \lambda = c(\lambda + \rho_G) - \rho_H$. This result associates to any irreducible G-representation a multiplet of H-representations, all of which share certain properties. For example, the alternating sum of the dimensions of the representations in the multiplet vanishes, and the value of the Casimir operator Δ on any of these representations is

$$\Delta_{c \bullet \lambda} = \left(\|c \bullet \lambda + \rho_H\|^2 - \|\rho_H\|^2 \right) = \left(\|\lambda + \rho_G\|^2 - \|\rho_H\|^2 \right).$$

By taking the index of the corresponding Dirac operator, we have a natural way to invert this process, taking an H-representation to its associated G-representation. Applying Theorem 3, we obtain

$$\operatorname{Index}_G \partial\!\!\!/^* \otimes U_{c \bullet \lambda} = (-1)^{c^{-1}} \, V_{c^{-1}(c \bullet \lambda + \rho_H) - \rho_G} = (-1)^c \, V_\lambda. \tag{1.19}$$

To see this result more directly, we can use Theorem 2 and the Peter-Weyl theorem to write this index as

$$\begin{aligned} \operatorname{Index}_G \partial\!\!\!/^* \otimes U_{c \bullet \lambda} &= \widehat{\bigoplus}_\lambda V_\lambda \otimes \left(V_\lambda^* \otimes (\mathbb{S}^+ - \mathbb{S}^-)^* \otimes U_{c \bullet \lambda} \right)^{\mathfrak{h}} \\ &= \widehat{\bigoplus}_\lambda V_\lambda \otimes \operatorname{Hom}_{\mathfrak{h}} \left(V_\lambda \otimes (\mathbb{S}^+ - \mathbb{S}^-), U_{c \bullet \lambda} \right), \end{aligned}$$

and then (1.19) follows immediately from (1.18). We also note that if we combine the equations (1.19) and (1.18), we get another identity similar to Bott's identity (1.17),

$$\sum_{w \in W_G / W_H} (-1)^w \, U_{w(\mu + \rho_H) - \rho_H} = (\mathbb{S}^+ - \mathbb{S}^-) \otimes \operatorname{Index}_G \partial\!\!\!/^* \otimes U_\mu.$$

Here we no longer have an explicit ρ_G-shift, since that comes automatically from taking the index of the Dirac operator. Going in the other direction, we have

$$\operatorname{Index}_G \partial\!\!\!/^* \otimes \left(V_\lambda \otimes (\mathbb{S}^+ - \mathbb{S}^-) \right) = (\#C) \, V_\lambda = \chi(G/H) \, V_\lambda,$$

where $\chi(G/H)$ is the Euler characteristic of G/H. In this case, we have

$$(\mathbb{S}^+ - \mathbb{S}^-)^* \otimes (\mathbb{S}^+ - \mathbb{S}^-) \cong \left(\Lambda^{\text{even}}(\mathfrak{p}) - \Lambda^{\text{odd}}(\mathfrak{p}) \right) \otimes \mathbb{C},$$

and so the Dirac operator becomes the de Rham complex tensored with V_λ.

CHAPTER 2

Loop Groups

2.1. Positive Energy Representations

Let G be a compact Lie group and let LG be the group of free loops on G, i.e., smooth maps from the circle S^1 to G, where the product of two loops is given by taking their pointwise product in G. The Lie algebra of a loop group is $L\mathfrak{g}$, the space of loops on the Lie algebra $\mathfrak{g}$. If we let S^1 act on LG by rotating the loops, this circle action induces a Fourier decomposition of the complexified Lie algebra $L\mathfrak{g}_{\mathbb{C}}$ as $\widehat{\bigoplus}_k \mathfrak{g}_{\mathbb{C}} z^k$, where Xz^n for $X \in \mathfrak{g}_{\mathbb{C}}$ denotes the loop $z \mapsto Xz^n$. We are interested in the representations of LG which intertwine with this S^1-action, or in other words we consider representations of the semi-direct product $S^1 \ltimes LG$. Such a representation $\mathcal{E}$ then decomposes into eigenspaces $\widehat{\bigoplus}_k \mathcal{E}(k)$, where the infinitesimal generator ∂_θ of the rotations acts by ik on $\mathcal{E}(k)$. We call this S^1 weight k the *energy*.

The representations that we want to work with are actually projective representations of LG. To realize them as genuine representations, we must introduce a central extension $\tilde{L}G$ of LG by S^1. This is analogous to taking the universal cover of a compact Lie group, except that here we must use a circle bundle $\tilde{L}G$ over LG instead of a finite cover. The corresponding central extension of the Lie algebra $L\mathfrak{g}$ is $\tilde{L}\mathfrak{g} \cong L\mathfrak{g} \oplus I\mathbb{R}$, where I is the infinitesimal generator of the central S^1 subgroup, and the Lie bracket of two loops $Xz^n \in L\mathfrak{g}_{\mathbb{C}}(n)$ and $Yz^m \in L\mathfrak{g}_{\mathbb{C}}(m)$ is given by

$$[Xz^n, Yz^m] = [X, Y]z^{n+m} - in\delta_{n,-m}\langle X, Y\rangle I, \tag{2.1}$$

where $\langle\ ,\ \rangle$ is an Ad-invariant inner product on the Lie algebra $\mathfrak{g}$. The central extension depends on the choice of inner product. However, if G is simply connected and simple, then there is a *universal central extension*, from which all other central extensions can be obtained as quotients. For a compact, simple Lie group, the Ad-invariant inner product is unique up to scaling, and the universal central extension corresponds to the smallest possible scaling. This smallest inner product is the *basic inner product*, scaled so that the highest root $\alpha_{\max}$ has $\|\alpha_{\max}\|^2 = 2$. To simplify matters, from here on we assume that G is simply connected and simple, we use the basic inner product on $\mathfrak{g}$, and we let $\tilde{L}G$ denote the universal central extension. However, with a few

modifications, the following discussion can be generalized to the case of an arbitrary compact Lie group.

Remark. If G is simply connected and simple, then we have $\pi_1(G) = \pi_2(G) = 1$ and $\pi_3(G) \cong \mathbb{Z}$. Taking the loop space shifts the homotopy down by one dimension, so we obtain $\pi_0(LG) = \pi_1(LG) = 1$ and $\pi_2(LG) \cong \mathbb{Z}$. We claim that taking the universal central extension $\tilde{L}G$ kills this π_2 term but otherwise leaves the homotopy of LG unchanged. From the homotopy exact sequence of the fibration $S^1 \to \tilde{L}G \to LG$ we see that $\pi_i(\tilde{L}G) \cong \pi_i(LG)$ for $i \geq 3$. To obtain π_1 and π_2, we consider the cohomology spectral sequence. In cohomology, we have $H^1(LG;\mathbb{Z}) = 0$ and $H^2(LG;\mathbb{Z}) \cong \mathbb{Z}$, and the spectral sequence gives us the exact sequence

$$0 \to H^1(\tilde{L}G;\mathbb{Z}) \to H^1(S^1;\mathbb{Z}) \xrightarrow{d_2} H^2(LG;\mathbb{Z}) \to H^2(\tilde{L}G;\mathbb{Z}) \to 0.$$

Now the image of the generator of $H^1(S^1;\mathbb{Z}) \cong \mathbb{Z}$ under the map d_2 is the first Chern class $c_1(\tilde{L}G) \in H^2(LG;\mathbb{Z})$. The universal central extension $\tilde{L}G$ corresponds to the circle bundle for which c_1 is the generator of $H^2(LG;\mathbb{Z}) \cong \mathbb{Z}$. The map d_2 is therefore an isomorphism, and the exact sequence tells us that $H^1(\tilde{L}G;\mathbb{Z}) = H^2(\tilde{L}G;\mathbb{Z}) = 0$. In terms of homotopy, we then obtain $\pi_1(\tilde{L}G) = \pi_2(\tilde{L}G) = 1$, as we claimed.

The action of S^1 on LG by rotation of loops extends to an action on $\tilde{L}G$, so we may consider representations of the semi-direct product $S^1 \ltimes \tilde{L}G$. From here on, we refer to these representations as representations of LG. Given such a representation, we call the weight of the central S^1 in $\tilde{L}G$ the *level* or *central charge*, and since this circle by definition commutes with the rest of the loop group, it follows that the level is constant on each irreducible representation.

Let T be a maximal torus of G. When considering the representation theory of loop groups, rather than taking the abelian subgroup LT as the maximal torus of LG, we instead use the maximal torus $S^1 \times T \times S^1$ of $S^1 \ltimes \tilde{L}G$. Here the first S^1 corresponds to rotation of loops, while the second comes from the central extension. The Cartan subalgebra is then $\mathbb{R} \oplus \mathfrak{t} \oplus \mathbb{R}$, and the weights of LG are of the form $\boldsymbol{\lambda} = (m, \lambda, h)$, where m is the energy, λ is a weight of G, and h is the level. In this notation, the root system Φ_{LG} of LG consists of the weights $(m, \alpha, 0)$ for $m \in \mathbb{Z}$ and $\alpha \in \Phi_G$, as well as the weights $(m, 0, 0)$ for all $m \neq 0$. Given a positive root system Φ_G^+ for G, we take the positive roots of LG to be the roots $(0, \alpha, 0)$ for $\alpha \in \Phi_G^+$, and $(m, \alpha, 0)$ for $m > 0$ and $\alpha \in \Phi_G$, as well as the roots $(m, 0, 0)$ for $m > 0$, counted with multiplicity $\dim \mathfrak{t}$.

We say that a representation $\mathcal{H}$ of LG is a *positive energy representation* if $\mathcal{H}(k) = 0$ for all $k < m$ for some fixed integer m, or in other words, if the constant energy subspaces have a minimum energy. In the literature, positive energy representations are often normalized so that this minimum energy is

0. However, for our purposes we will consider positive energy representations with the full spectrum of minimum energies. If we restrict our attention to the positive energy representations, we find that the representation theory of loop groups behaves quite analogously to the representation theory of compact Lie groups. In particular, the positive energy representations have the following fundamental properties (for a complete discussion, see [**6**]):

(i) A positive energy representation is *completely reducible* as a (possibly infinite) direct sum of irreducible positive energy representations.
(ii) The irreducible positive energy representations are of *finite type*; if $\mathcal{H}$ is an irreducible positive energy representation, then each of its constant energy subspaces $\mathcal{H}(k)$ is a finite dimensional representation of G.
(iii) Every irreducible positive energy representation $\mathcal{H}$ has a unique *lowest weight* $\boldsymbol{\lambda} = (m, \lambda, h)$, in the sense that $\boldsymbol{\lambda} - \boldsymbol{\alpha}$ is not a weight of $\mathcal{H}$ for any positive root $\boldsymbol{\alpha} \in \Phi^+_{LG}$. The lowest weight space is one dimensional and generates $\mathcal{H}$.
(iv) A weight (m, λ, h) is *anti-dominant* for LG if $\langle -\lambda, \alpha_i \rangle \geq 0$ for each of the simple roots α_i of G (i.e., $-\lambda$ is dominant for G) and $\langle -\lambda, \alpha_{\max} \rangle \leq h$, where $\alpha_{\max}$ is the highest root of G. The lowest weight of a positive energy representation is anti-dominant, and every anti-dominant weight is realized as the lowest weight of some positive energy representation.

As a consequence of (iii), we see that an irreducible positive energy representation $\mathcal{H}$ is completely characterized by its minimum energy m, its minimum energy subspace $\mathcal{H}(m) \cong V_{-\lambda}$, and its level h. From (iv) we conclude that for a positive energy representation, the level h is always non-negative and is zero only for the trivial representation. Also, we observe that if we fix the minimum energy m, then there are only finitely many positive energy representations at each level h.

2.2. The Dirac Operator on LG

The primary difficulty in understanding the Dirac operator on loop spaces is finding a suitable domain on which the operator acts. As a first step towards this goal, we examine the spin representation of a loop group. The adjoint representation of LG on its Lie algebra $L\mathfrak{g}$ extends naturally to a representation of $S^1 \ltimes \tilde{L}G$, where S^1 acts on $L\mathfrak{g}$ by rotating the loops, and the central extension acts trivially. At the Lie algebra level, the adjoint representation gives us a homomorphism $\mathrm{ad} : L\mathfrak{g} \to \mathfrak{so}(L\mathfrak{g})$, which in turn lifts to a spin representation $\widetilde{\mathrm{ad}} : L\mathfrak{g} \to \mathfrak{spin}(L\mathfrak{g})$. To see this explicitly, given any $X \in \mathfrak{g}_{\mathbb{C}}$, we let X^n denote the loop Xz^n. Letting $\{X_i\}$ be a basis for $\mathfrak{g}_{\mathbb{C}}$, we take $\{X_i^n\}$ to be a basis for $L\mathfrak{g}_{\mathbb{C}}$, and we use the L^2 inner product on $L\mathfrak{g}_{\mathbb{C}}$

given by $\langle X^n, Y^m\rangle = \langle X, Y\rangle \delta_{n,-m}$. Generalizing (1.3) to loop groups, we have

$$\widetilde{\mathrm{ad}}\,\xi = \frac{1}{4}\sum_{n\in\mathbb{Z},j} (X_j^*)^{-n}\cdot[\xi, X_j^n]$$

for any loop $\xi \in L\mathfrak{g}_{\mathbb{C}}$. The generator ∂_θ of rotations also lifts, and we obtain

$$\widetilde{\mathrm{ad}}\,\partial_\theta = \frac{1}{4}\sum_{n\in\mathbb{Z},j} (X_j^*)^{-n}\cdot\partial_\theta X_j^n = \frac{1}{4}\sum_{n\in\mathbb{Z},j} in\,(X_j^*)^{-n}\cdot X_j^n. \tag{2.2}$$

The image $\widetilde{\mathrm{ad}}\,I$ of the generator of the central extension is actually a central element in an extension of $\mathfrak{spin}(L\mathfrak{g})$, as we shall demonstrate below.

To obtain the spin representation for LG, we decompose its Lie algebra $L\mathfrak{g}_{\mathbb{C}}$ as $\mathfrak{g}_{\mathbb{C}} \oplus W \oplus \bar{W}$, where $W = \bigoplus_{k>0} \mathfrak{g}_{\mathbb{C}} z^k$ and its conjugate is $\bar{W} = \bigoplus_{k<0} \mathfrak{g}_{\mathbb{C}} z^k$. Give this polarization of $L\mathfrak{g}_{\mathbb{C}}$, we define the corresponding spin space to be

$$\mathcal{S} = \mathbb{S}_{\mathfrak{g}} \otimes \Lambda^*(W). \tag{2.3}$$

The Clifford action of $L\mathfrak{g}_{\mathbb{C}}$ on $\mathcal{S}$ is then given by

$$c(X) = \begin{cases} i\otimes\varepsilon(\xi) & \text{for } \xi\in W,\\ i\otimes\iota(\xi) & \text{for } \xi\in\bar{W},\\ c(X)\otimes(-1)^F & \text{for } X\in\mathfrak{g}_{\mathbb{C}},\end{cases}$$

where ε and ι denote exterior and interior multiplication respectively, and $(-1)^F$ takes the value $+1$ on $\Lambda^{\mathrm{even}}(W)$ and -1 on $\Lambda^{\mathrm{odd}}(W)$. Given an Ad-invariant inner product on $\mathfrak{g}$, we choose an orthonormal basis $\{H_j\}$ of $\mathfrak{t}$, and we pick generators $E_\alpha \in \mathfrak{g}_\alpha$ for each root $\alpha \in \Phi_G$ such that $\langle E_\alpha, E_{-\alpha}\rangle = \frac{1}{2}$. We note that since $[\widetilde{\mathrm{ad}}\,\xi, \eta] = (\mathrm{ad}\,\xi)(\eta)$ holds in the Clifford algebra, the actions of H_j^n and E_α^n on $\mathcal{S}$ shift the weights of vectors by the roots $(n, 0)$ and (n, α) respectively. Let v be a lowest weight vector of $\mathcal{S}$, which is characterized by

$$E_{-\alpha}v = 0 \text{ for } \alpha\in\Phi^+, \quad H_j^{-n}v = 0 \text{ for } n>0, \quad E_\alpha^{-n}v = 0 \text{ for } n>0, \alpha\in\Phi.$$

We now compute the weight of v. Given any $H \in \mathfrak{t}$, we have

$$\begin{aligned}(\widetilde{\mathrm{ad}}\,H)\,v &= \frac{1}{4}\Big(\sum_{j,n} H_j^{-n}\cdot[H, H_j^n] + \sum_{\alpha,n} 2E_{-\alpha}^{-n}\cdot[H, E_\alpha^n]\Big)v\\ &= \frac{1}{2}\sum_{\alpha\in\Phi^+} i\alpha(H)\,(E_{-\alpha}\cdot E_\alpha)\,v + \frac{1}{2}\sum_{n>0}\sum_{\alpha\in\Phi} i\alpha(H)\,\big(E_{-\alpha}^{-n}\cdot E_\alpha^n\big)\,v\\ &= -\frac{1}{2}\sum_{\alpha\in\Phi^+} i\alpha(H)\,v - \frac{1}{2}\sum_{n>0}\sum_{\alpha\in\Phi} i\alpha(H)\,v\\ &= -i\rho(H)\,v,\end{aligned}$$

where the sum of all the roots of G vanishes since roots come in $\pm\alpha$ pairs, and we are left with just the $n = 0$ contribution. The energy of the lowest weight vector v is then obtained by applying $\widetilde{\mathrm{ad}}\,\partial_\theta$ as given by (2.2):

$$\begin{aligned}(\widetilde{\mathrm{ad}}\,\partial_\theta)\,v &= \frac{1}{4}\sum_{n>0,j} in\left((X_j^*)^{-n}\cdot X_j^n\right)v \\ &= -\frac{1}{2}\sum_{n>0,j} inv = -\frac{1}{2}\,i\zeta(-1)\dim\mathfrak{g}\,v = i\,\frac{1}{24}\dim\mathfrak{g}\,v,\end{aligned} \tag{2.4}$$

where we use the identity $\sum_{n>0} n = \zeta(-1) = -\frac{1}{12}$, which is obtained by analytic continuation of the Riemann zeta function $\zeta(s) = \sum_{n>0} n^{-s}$.

Remark. Such infinite sums are common in quantum field theory, and much of the theory is devoted to making sense of them. These infinite constants are often simply ignored or avoided by taking normal ordered operators as discussed later in this section. For our purposes, we can always shift the minimum energy by a constant by multiplying the spin representation by a character of S^1, and this technique works, at least formally, even if that constant is infinite. In particular, the following discussion is unchanged if we normalize the minimum energy to be 0. On the other hand, the Riemann zeta function identity gives us a way to carefully keep track of the infinite terms, and in string theory, this $\zeta(-1)$ term has a physical interpretation as giving fermions a non-zero vacuum energy. We also note that the constant 24 that appears in (2.4) and throughout string theory is related to the 26 dimensions required for the bosonic string; a one dimensional string sweeps out a two dimensional world sheet as it evolves over time, and this world sheet in turn can be interpreted as living in a 24 dimensional space.

We recall from Section 1.2 that the highest weight of the spin representation of a compact Lie group is half the sum of the positive roots. Noting that

$$\begin{aligned}-\frac{1}{2}\sum_{\boldsymbol{\alpha}\in\Phi_{LG}^+}\boldsymbol{\alpha} &= -\frac{1}{2}\sum_{\alpha\in\Phi_G^+}(0,\alpha) - \frac{1}{2}\sum_{m>0,\alpha\in\Phi_G}(m,\alpha) - \frac{1}{2}\sum_{m>0}\dim\mathfrak{t}\,(m,0) \\ &= \Big(-\frac{1}{2}\dim\mathfrak{g}\sum_{m>0} m, -\rho\Big),\end{aligned}$$

we see that the lowest weight of the spin representation $\mathcal{S}$, viewed as a representation of the semi-direct product $S^1 \ltimes LG$, is likewise half the sum of the negative roots of $S^1 \ltimes LG$.

In the loop group case, we must also consider the level of the spin representation. The simplest way to obtain the generator I of the central extension is by choosing any $H \in \mathfrak{t}$ normalized so that $\|H\|^2 = 1$ and noting that by

(2.1), we have the commutator $[H^1, H^{-1}] = -iI$. Applying $\widetilde{\mathrm{ad}}$, we obtain

$$\widetilde{\mathrm{ad}}\, I = i\,[\widetilde{\mathrm{ad}}\, H^1, \widetilde{\mathrm{ad}}\, H^{-1}].$$

For a general $n \neq 0$, we compute

$$\begin{aligned}\widetilde{\mathrm{ad}}\, H^n &= \frac{1}{2} \sum_{\alpha \in \Phi^+, m} i\alpha(H) \left(E_{-\alpha}^{-m} \cdot E_\alpha^{n+m} - E_\alpha^{n+m} \cdot E_{-\alpha}^{-m}\right) \\ &= - \sum_{\alpha \in \Phi^+, m} i\alpha(H) E_\alpha^{n+m} \cdot E_{-\alpha}^{-m}.\end{aligned}$$

We can now use this to determine the level of v:

$$\begin{aligned}(\widetilde{\mathrm{ad}}\, I)\, v &= -i \sum_{\alpha, \beta \in \Phi^+, m, k} \alpha(H)\beta(H) \left[E_\alpha^{m+1} \cdot E_{-\alpha}^{-m}, E_\beta^{k-1} \cdot E_{-\beta}^{-k}\right] v \\ &= -i \sum_{\alpha \in \Phi^+, m} \alpha(H)^2 \left[E_\alpha^{m+1} \cdot E_{-\alpha}^{-m}, E_\alpha^{m} \cdot E_{-\alpha}^{-m-1}\right] v \\ &= i \sum_{\alpha \in \Phi^+, m} \alpha(H)^2 \left(E_\alpha^{m+1} \cdot E_{-\alpha}^{-m-1} - E_\alpha^{m} \cdot E_{-\alpha}^{-m}\right) v \\ &= i \sum_{\alpha \in \Phi^+} \alpha(H)^2 v.\end{aligned}$$

For the final step in the above computation, we note that

$$\left(E_\alpha^{m+1} \cdot E_{-\alpha}^{-m-1} - E_\alpha^{m} \cdot E_{-\alpha}^{-m}\right) v = \begin{cases} (0+0)\, v = 0 & \text{for } m > -1, \\ (0+1)\, v = v & \text{for } m = -1, \\ (-1+1)\, v = 0 & \text{for } m < -1. \end{cases}$$

Putting together the above weights, we can completely characterize the spin representation $\mathcal{S}$ in terms of irreducible positive energy representations of LG.

THEOREM 4. *The spin representation $\mathcal{S}$ of LG is a direct sum of multiple copies of the positive energy represention with lowest weight $(m_G, -\rho_G, c_G)$, where the minimum energy is $m_G = \frac{1}{24} \dim \mathfrak{g}$, the weight ρ_G is half the sum of the positive roots of G, and the level is*

$$c_G = \langle \alpha_{\max}, \rho_G \rangle + 1,$$

where $\alpha_{\max}$ is the highest root of G.

PROOF. All of the weights come directly from our computations above, with the exception of the level c_G. Given $H \in \mathfrak{t}$ with $\|H\|^2 = 1$, we have

$$\begin{aligned} c_G &= \sum_{\alpha\in\Phi^+} \alpha(H)^2 = \frac{1}{2}\sum_{\alpha\in\Phi} \alpha(H)^2 = -\frac{1}{2}\sum_{\alpha\in\Phi} \langle [H,[H,E_\alpha]], E_\alpha^*\rangle \\ &= -\frac{1}{2}\sum_{\alpha\in\Phi} \langle (\operatorname{ad} E_\alpha^*)(\operatorname{ad} E_\alpha)H, H\rangle = \frac{1}{2}\langle \Delta_{\mathrm{ad}} H, H\rangle \\ &= \frac{1}{2}\left(\|\alpha_{\max}+\rho\|^2 - \|\rho\|^2\right) = \langle \alpha_{\max}, \rho\rangle + 1, \end{aligned}$$

recalling that we are using the basic inner product satisfying $\|\alpha_{\max}\|^2 = 2$. $\square$

Since loop spaces are infinite dimensional, the notion of integrating a function over the entire loop space does not make sense (unless the reader is accustomed to the formalism of path integrals). As a result, when considering differential operators on a loop space, there is no space of L^2 functions to work with. However, if our loop space is actually a loop group, we can build a suitable Hilbert space of functions out of the positive energy representations. Using the Peter-Weyl decomposition of L^2 functions on a compact Lie group as our model, we define

$$L^2(\tilde{L}G) := \widehat{\bigoplus}_{\lambda,h} \mathcal{H}_{\lambda,h} \otimes \mathcal{H}^*_{\lambda,h}, \tag{2.5}$$

where $\mathcal{H}_{\lambda,h}$ is the positive energy representation with lowest weight (λ, h) and minimum energy 0, and its adjoint representation $\mathcal{H}^*_{\lambda,h}$ is a negative energy representation of level $-h$. Since we are interested in sections over the loop group LG itself, rather than its central extension $\tilde{L}G$, we view $\tilde{L}G$ as a principal S^1 bundle over LG, and we introduce the *fundamental line bundle*

$$\mathcal{L} = \tilde{L}G \times_{S^1} \mathbb{C}$$

over LG, where S^1 acts on $\mathbb{C}$ by complex rotation. We can then take

$$L^2(\mathcal{L}^h) = \left(L^2(\tilde{L}G) \otimes \mathbb{C}_h\right)^{S^1} = \widehat{\bigoplus}_{\lambda} \mathcal{H}_{\lambda,h} \otimes \mathcal{H}^*_{\lambda,h} \tag{2.6}$$

to be the Hilbert space of sections of $\mathcal{L}^h$.

Generalizing the twisted Dirac operator (1.15), we note that we want to use spinors taking values in the adjoint $\mathcal{S}^*$ of our spin representation $\mathcal{S}$. In the finite dimensional case, we saw that the spin representation is self-adjoint, and that taking its adjoint simply fixed or interchanged the two half-spin representations, depending on the dimension. In the loop group case, the adjoint spin representation $\mathcal{S}^*$ is actually a negative energy representation of level $-c_G$, and taking the adjoint now makes a significant difference to the

theory. For our Hilbert space of spinors, we take

$$L^2(\mathcal{L}^{h+c_G} \otimes \mathcal{S}^*) = \widehat{\bigoplus}_\lambda \mathcal{H}_{\lambda,h} \otimes \big(\mathcal{H}_{\lambda,h} \otimes \mathcal{S}\big)^*. \tag{2.7}$$

Here, the tensor product $\mathcal{H}_{\lambda,h} \otimes \mathcal{S}$ is again a positive energy representation of finite type, and its level is $h + c_G$.

As a formal element of $U(\tilde{L}\mathfrak{g}) \otimes \mathrm{Cl}(L\mathfrak{g})$, the Dirac operator on LG is

$$\not{\partial} = \sum_{n\in\mathbb{Z},j} \big(r(X_j^n) \otimes X_j^{-n} + q \otimes X_j^{-n} \cdot \widetilde{\mathrm{ad}}\, X_j^n\big), \tag{2.8}$$

where the X_i are an orthonormal basis of $\mathfrak{g}$, and the constant q is a "charge". Taking $q = \frac{1}{2}$ gives us the Dirac operator coming from the Levi-Civita connection, while $q = \frac{1}{3}$ gives the loop group analogue of Kostant's Dirac operator. The right action of LG on spinors corresponds to the diagonal representation $\varrho : \tilde{L}\mathfrak{g} \to U(\tilde{L}\mathfrak{g}) \otimes \mathrm{Cl}(L\mathfrak{g})$ given by

$$\varrho = r \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\,.$$

Unlike in the finite dimensional case, we find that the Dirac operator on LG does not actually commute with ϱ. Computing the commutator $[\varrho(Y^m), \not{\partial}]$ as in the proof of Lemma 3, we obtain extra terms from the central extension:

$$\begin{aligned}\Big[\varrho(Y^m), \not{\partial}\Big] &= \sum_{j,n} -im\delta_{m,-n}\langle Y, X_j\rangle \big(r(I) \otimes X_j^{-n} + q \otimes X_i^{-n} \cdot \widetilde{\mathrm{ad}}\, I\big) \\ &= -im\big(r(I) \otimes 1 + q \otimes \widetilde{\mathrm{ad}}\, I\big)\big(1 \otimes Y^m\big).\end{aligned}$$

On the other hand, the Dirac operator automatically commutes with $\varrho(I)$ since I is in the center of $\tilde{L}\mathfrak{g}$, and it also commutes with the energy operator $\varrho(\partial_\theta)$:

$$\begin{aligned}[\varrho(\partial_\theta), \not{\partial}] = \sum_{j,n}\Big(&r(\partial_\theta X_j^n) \otimes X_j^{-n} + r(X_j^{-n}) \otimes (\partial_\theta X_j^{-n}) \\ &+ q \otimes (\partial_\theta X_j^{-n}) \cdot \widetilde{\mathrm{ad}}\, X_j^n + q \otimes X_j^{-n} \cdot \widetilde{\mathrm{ad}}(\partial_\theta X_j^n)\Big) = 0,\end{aligned} \tag{2.9}$$

since $\partial_\theta X_i^n = inX_i^n$. The Dirac operator therefore commutes with the diagonal action ϱ only for the zero energy subgroup $S^1 \times G \times S^1$ of $S^1 \ltimes \tilde{L}G$.

To find the square of the Dirac operator on LG, we perform the same calculations as in Section 1.2. The only difference here is the addition of terms coming from the central extension of LG. We find that the square of the first

term in the Dirac operator is

$$\begin{aligned}\Big(\sum_{j,n} r(X_j^n)\otimes X_j^{-n}\Big)^2 &= -\sum_{j,n} r(X_j^n)r(X_j^{-n})\otimes 1 + 2\sum_{j,n} r(X_j^n)\otimes \widetilde{\mathrm{ad}}\, X_j^{-n} \\ &\quad + \frac{1}{2}\sum_{j,k,n,m} r(-in\delta_{n,-m}\langle X_j, X_k\rangle I)\otimes X_j^{-n}\cdot X_k^{-m} \\ &= 2\Delta_r\otimes 1 - \Delta_\varrho + 1\otimes \Delta_{\widetilde{\mathrm{ad}}} - r(I)\otimes \frac{1}{2}\sum_{j,n} inX_j^{-n}\cdot X_j^n,\end{aligned}$$

where we note that the Casimir operator for $\varrho = r\otimes 1 + 1\otimes \widetilde{\mathrm{ad}}$ is

$$\Delta_\varrho = \Delta_r\otimes 1 - 2\sum_{j,n} r(X_j^n)\otimes \widetilde{\mathrm{ad}}\, X_j^{-n} + 1\otimes \Delta_{\widetilde{\mathrm{ad}}}.$$

The cross terms in the square of the Dirac operator are not affected by the central extension:

$$\begin{aligned}\Big\{\sum_{j,n} r(X_j^n)\otimes X_j^{-n}, q\otimes \sum_{k,m} X_k^{-m}\cdot \widetilde{\mathrm{ad}}\, X_k^m\Big\} &= -6q\sum_{j,n} r(X_j^n)\otimes \widetilde{\mathrm{ad}}\, X_j^{-n} \\ &= 3q\left(\Delta_\varrho - \Delta_r\otimes 1 - 1\otimes \Delta_{\widetilde{\mathrm{ad}}}\right).\end{aligned}$$

Finally, the square of the cubic term in the Dirac operator is

$$\begin{aligned}\Big(q\otimes \sum_{j,n} X_j^{-n}\cdot \widetilde{\mathrm{ad}}\, X_j^n\Big)^2 &= 3q^2\otimes\Big(-\sum_{j,n} \widetilde{\mathrm{ad}}(X_j^n)\cdot \widetilde{\mathrm{ad}}(X_j^{-n}) \\ &\quad + \frac{1}{2}\sum_{j,k,n,m} X_j^{-n}\cdot X_k^{-m}\cdot \widetilde{\mathrm{ad}}(-in\delta_{n,-m}\langle X_j, X_k\rangle I)\Big) \\ &= 3q^2\otimes \Delta_{\widetilde{\mathrm{ad}}} - 3q^2\otimes \widetilde{\mathrm{ad}}(I)\cdot \frac{1}{2}\sum_{j,n} inX_j^{-n}\cdot X_j^n.\end{aligned}$$

Combining these three terms, and recalling the expression (2.2) for $\widetilde{\mathrm{ad}}\,\partial_\theta$, we obtain the following Weitzenböck formula for the square of the Dirac operator:

$$\begin{aligned}\not{\partial}^2 &= (2-3q)\Delta_r\otimes 1 - (1-3q)\Delta_\varrho + (1-3q+3q^2)\otimes \Delta_{\widetilde{\mathrm{ad}}} \\ &\quad - \big(r(I)\otimes 1 + 3q^2\otimes \widetilde{\mathrm{ad}}\, I\big)\big(1\otimes 2\,\widetilde{\mathrm{ad}}\,\partial_\theta\big).\end{aligned}$$

Remark. The Casimir operator on a loop group, like the Dirac operator, commutes only with the $S^1\times G\times S^1$ subgroup of $S^1\ltimes \tilde{L}G$. However, we can modify the Casimir operator slightly so that it does indeed commute with the

whole group. Working in $U(\tilde{L}\mathfrak{g})$, the commutator with a loop $Y^n \in L\mathfrak{g}_{\mathbb{C}}$ is

$$\begin{aligned}[Y^m, \Delta] &= -\sum_{j,n} \left([Y^m, X_j^n]X_j^{-n} + X_j^n[Y^m, X_j^{-n}]\right) \\ &= \sum_{j,n} im\langle Y, X_j\rangle \left(\delta_{m,-n} I X_j^{-n} + \delta_{m,n} X_j^n I\right) \\ &= 2imIY^m = 2I(\partial_\theta Y^m).\end{aligned}$$

Adding a term to cancel this commutator, we find that the operator

$$\tilde{\Delta} = \Delta + 2I\partial_\theta$$

commutes with the action of the entire group $S^1 \ltimes \tilde{L}G$. We note that this extended Casimir operator $\tilde{\Delta}$ is actually the Casimir operator for the extended loop group $S^1 \ltimes \tilde{L}G$, using the Ad-invariant inner product on its Lie algebra $\mathbb{R} \tilde{\oplus} \tilde{L}\mathfrak{g}$ given in [**6**, §4.9].

When working with positive energy representations, we require that our operators be normal ordered, with the energy-lowering X_i^{-n} terms placed to the right of the energy-raising X_i^n terms. Since a positive energy representation has a minimum energy, we can then sum over all energies $n \in \mathbb{Z}$ and the total operator will converge. As we saw in (2.4), we define $(\widetilde{\mathrm{ad}}\,\partial_\theta)^+$ as:

$$\begin{aligned}(\widetilde{\mathrm{ad}}\,\partial_\theta)^+ :&= -\frac{1}{2}\sum_{n>0,j} inX_j^n \cdot X_j^{-n} \\ &= \frac{1}{4}\sum_{n,j} inX_j^{-n} \cdot X_j^n + \frac{1}{2}\sum_{n>0,j} in \\ &= \widetilde{\mathrm{ad}}\,\partial_\theta + \frac{1}{2}i\zeta(-1)\dim\mathfrak{g}.\end{aligned}$$

Likewise, we can define a normal ordered Casimir operator Δ^+ by:

$$\begin{aligned}\Delta^+ :&= -\sum_j (X_j)^2 - 2\sum_{n>0,j} X_j^n X_j^{-n} \\ &= -\sum_{n\in\mathbb{Z},j} X_j^n X_j^{-n} + \sum_{n>0,j} inI \\ &= \Delta + i\zeta(-1)\dim\mathfrak{g}\, I.\end{aligned}$$

Writing our Dirac operator in normal ordered form, we observe that all of the infinite $\zeta(-1)$ terms cancel, and we obtain

$$\begin{aligned}\partial\!\!\!/^2 = (2-3q)\Delta_r^+ \otimes 1 - (1-3q)\Delta_\varrho^+ + (1-3q+3q^2)\otimes\Delta_{\widetilde{\mathrm{ad}}}^+ \\ - \left(r(I)\otimes 1 + 3q^2 \otimes \widetilde{\mathrm{ad}}\, I\right)\left(1 \otimes 2(\widetilde{\mathrm{ad}}\,\partial_\theta)^+\right).\end{aligned}$$

Although the Casimir operator Δ^+ itself does not commute with the whole loop group LG, we saw above that by adding terms to cancel the commutators, we can obtain an operator that does commute with LG and is thus constant on each irreducible representation. This extended normal ordered Casimir operator is then

$$\tilde{\Delta}^+ = \Delta^+ + 2(I + ic_G)\partial_\theta,$$

where the $2ic_G\partial_\theta$ term comes from taking the normal ordering (see [**6**, §9.4] for the details). Using this to determine the action of Δ^+ on the irreducible positive energy repesentation $\mathcal{H}_{m,\lambda,h}$ with lowest weight (m, λ, h), we obtain

$$\Delta^+_{m,\lambda,h} = \|\lambda - \rho\|^2 - \|\rho\|^2 + 2(h + c_G)(-i\partial_\theta - m). \tag{2.10}$$

In other words, on the minimum energy subspace, the Casimir operators for the loop group LG and the Lie group G agree, while raising the energy by k increases the eigenvalue of the Casimir operator by $2(h + c_G)k$.

Using our expression (2.10) for the Casimir operator on a positive energy representation, we can now write down the square of the Dirac operator acting on $\mathcal{H}_{m,\lambda,h} \otimes \mathcal{S}$. On the irreducible subrepresentation $\mathcal{H}_{k,\mu,h+c_G}$, we obtain

$$\begin{aligned}\partial\!\!\!/^2 = {} & (2-3q)\|\lambda-\rho\|^2 - (1-3q)\|\mu-\rho\|^2 + (2-3q)(1-3q)\|\rho\|^2 \\ & + (2-3q)2(h+c_G)(E_r - m) - (1-3q)2(h+2c_G)(E_r + E_{\tilde{\mathrm{ad}}} - k) \\ & + (2h + (2-6q+9q^2)2c_G)(E_{\tilde{\mathrm{ad}}} - m_G),\end{aligned}$$

where $E = -i\partial_\theta$ gives the energy of the eigenspace. Taking $q = \frac{1}{3}$ to obtain the loop group version of Kostant's Dirac operator, our Weitzenböck formula takes the form:

$$\partial\!\!\!/^2_{1/3} = \|\lambda - \rho\|^2 + 2(h + c_G)(E_r + E_{\tilde{\mathrm{ad}}} - m - m_G).$$

Note that the square of the Dirac operator takes on a constant value on each of the constant energy subspaces. Since the Dirac operator itself commutes with $S^1 \times G \times S^1$, its spectrum on the constant energy subspaces is given, up to sign, by taking the square root of the eigenvalues of $\partial\!\!\!/^2$.

2.3. The Infinite Central Charge Limit

In [**7**], Taubes considers the Dirac operator on the loop space LM, where M is a spin manifold. Restricting to an infinitesimal neighborhood of the constant loops $M \subset LM$, Taubes constructs a local Dirac operator on the normal bundle $\mathcal{N}M$ of M in LM. If G is a Lie group, then the construction in the previous section gives a global Dirac operator on LG. In this section, we show that in the infinite central charge limit, our global Dirac operator coincides with the local Dirac operator on the normal bundle to G.

We recall from Section 2.1 that a weight (λ, h) is anti-dominant for $\tilde{L}G$ if and only if $-\lambda$ is a dominant weight for G and $\langle -\lambda, \alpha_{\max}\rangle \leq h$. If we let the level or central charge h tend to infinity, then this second condition is automatically satisfied, and the anti-dominant weights of $\tilde{L}G$ correspond precisely to the dominant weights of G.

If ρ is a representation of $L\mathfrak{g}$ of level h, then its commutators are given by

$$[\rho(X^n), \rho(Y^m)] = \rho([X,Y]^{n+m}) + hn\delta_{n,-m}\langle X, Y\rangle.$$

As h increases, the central charge term dominates the Lie algebra bracket. Rescaling the representation, we take $\tilde{\rho} = h^{-\frac{1}{2}}\rho$, and the commutators then become

$$[\tilde{\rho}(X^n), \tilde{\rho}(Y^m)] = h^{-\frac{1}{2}}\tilde{\rho}([X,Y]^{n+m}) + n\delta_{n,-m}\langle X, Y\rangle.$$

Now, taking the limit as h tends to infinity, the term coming from the bracket on $L\mathfrak{g}$ vanishes, and we are left with just

$$[\tilde{\rho}(X^n), \tilde{\rho}(Y^m)] = n\delta_{n,-m}\langle X, Y\rangle. \tag{2.11}$$

However, we do not lose all of the Lie algebra structure. The Lie algebra $\mathfrak{g}$ of constant loops, viewed as the energy zero subspace in $L\mathfrak{g}$, does not need to be rescaled, and regardless of the scaling we have

$$[\rho(X), \tilde{\rho}(Y^m)] = \tilde{\rho}([X,Y]^m). \tag{2.12}$$

These commutators are therefore preserved as the level h tends to infinity. Letting $\hat{\mathfrak{g}}$ denote the abelian Lie algebra sharing the same underlying vector space as $\mathfrak{g}$, we observe that the commutators (2.11) are the brackets for the Lie algebra $\tilde{L}\hat{\mathfrak{g}}$. The constant loops $\hat{\mathfrak{g}}$ are in the center of $\tilde{L}\hat{\mathfrak{g}}$ and thus decouple from the action, leaving us with an action of the Lie algebra $\tilde{L}\hat{\mathfrak{g}}/\hat{\mathfrak{g}}$. These constant loops are replaced by the action of $\mathfrak{g}$, which intertwines with the action of $\tilde{L}\hat{\mathfrak{g}}$ according to the commutators (2.12). The infinite central charge representations of $\tilde{L}\mathfrak{g}$ therefore correspond to the representations of the semi-direct sum $\mathfrak{g} \tilde{\oplus} (\tilde{L}\hat{\mathfrak{g}}/\hat{\mathfrak{g}})$.

Let $\mathcal{H}$ be an irreducible positive energy representation of $\tilde{L}\mathfrak{g}$ with infinite central charge. If m is the minimum energy of $\mathcal{H}$, then the minimum energy subspace $V = \mathcal{H}(m)$ is an irreducible finite dimensional representation of G. For any vector $v \in V$, we then have $\tilde{\rho}(X^{-n})v = 0$ for all $n < 0$. For the positive energy operators, we write $W = \bigoplus_{n>0} \mathfrak{g}_{\mathbb{C}} z^n$, and we let

$$S^*(W) = \bigoplus_{k\geq 0} S^k(W) = \bigotimes_{n>0} S^*(\mathfrak{g}_{\mathbb{C}} z^n)$$

denote the symmetric algebra on W, i.e., the space of all polynomials in the elements of W. Noting that $\tilde{\rho}(X^n)$ and $\tilde{\rho}(Y^m)$ commute if both $n, m > 0$, we

see that the the linear map given by

$$v \otimes (X_1^{n_1} \cdots X_k^{n_k}) \mapsto \tilde{\rho}(X_1^{n_1}) \cdots \tilde{\rho}(X_k^{n_k}) v$$

is a vector space homomorphism $V \otimes S^*(W) \to \mathcal{H}$. Furthermore, we note that this homomorphism is equivariant with respect to the $L\mathfrak{g}$-action on $V \otimes S^*(W)$ given by

$$\begin{aligned} X &\mapsto \rho(X) \otimes 1 + 1 \otimes \operatorname{ad} X && \text{for} \quad X \in \mathfrak{g}, \\ X^n &\mapsto 1 \otimes X^n && \text{for} \quad X^n \in W, \\ X^{-n} &\mapsto 1 \otimes -n\partial_{X^n} && \text{for} \quad X^{-n} \in \bar{W}, \end{aligned}$$

and since $\mathcal{H}$ is an irreducible representation of $L\mathfrak{g}$, we therefore conclude that $\mathcal{H} \cong V \otimes S^*(W)$. The irreducible positive energy representations of $\tilde{L}\mathfrak{g}$ with infinite central charge are thus the representations $V_\lambda \otimes S^*(W)$, where V_λ is the irreducible representation of G with lowest weight λ.

Using this description of the infinite central charge representations, the Hilbert space of sections of the line bundle $\mathcal{L}^h$ given by (2.6) becomes

$$\begin{aligned} L^2(\mathcal{L}^\infty) &= \widehat{\bigoplus_\lambda} \big(V_\lambda \otimes S^*(W)\big) \otimes \big(V_\lambda \otimes S^*(W)\big)^* \\ &= L^2(G) \otimes \big(S^*(W) \otimes S^*(\bar{W})\big) \end{aligned}$$

in the limit as the level h tends to infinity. In other words, we may view the sections of the line bundle $\mathcal{L}^\infty$ over LG as sections of an infinite dimensional bundle $\mathcal{B}$ over G with fibers

$$\mathcal{B}_g = S^*(W) \otimes S^*(\bar{W}) = S^*(W \oplus \bar{W}) \cong S^*(W \otimes \mathbb{C}).$$

If we let $\mathcal{N}G$ be the normal bundle to G embedded in LG as the constant loops, we note that the fibers of $\mathcal{N}G$ have a natural complex structure and that we can identify the vector space $W = \bigoplus_{n>0} \mathfrak{g}_\mathbb{C} z^n$ with a dense subspace of each fiber. The complex dual $W^* \otimes \mathbb{C}$ is the space of all linear complex-valued functions on W, and the identity $W^* \otimes \mathbb{C} \cong W^* \otimes \bar{W}^*$ lets us decompose those functions into their holomorphic and anti-holomorphic parts. The symmetric algebra $S^*(W^* \otimes \mathbb{C})$ then consists of all complex-valued polynomial functions on W. Identifying W^* with $\bar{W}$ via our inner product on $L\mathfrak{g}_\mathbb{C}$, we see that the sections of the bundle $\mathcal{B}$ over G are functions on $\mathcal{N}G$ which are polynomials along each fiber. The $S^*(W) \cong S^*(\bar{W}^*)$ component corresponds to the anti-holomorphic polynomials, while $S^*(\bar{W}) \cong S^*(W^*)$ corresponds to the holomorphic polynomials.

Letting $\{X_j\}$ be an orthonormal basis for $\mathfrak{g}$, we take $\{X_j^n\}$ for $n > 0$ as a complex basis for W. We let x_j^n for $n > 0$ be the corresponding complex coordinates on the fibers of $\mathcal{N}G$, and we write $\bar{x}_j^n$ for their complex conjugates. The fibers of $\mathcal{B}$ then consist of all polynomials in the variables x_j^n and $\bar{x}_j^n$.

Letting l and r be the left and right actions of $L\mathfrak{g}$ on sections of $\mathcal{L}^\infty$, we see that in this context l and r are now actions on the anti-holomorphic and holomorphic polynomials, respectively. For any $n > 0$, these actions are given by

$$l(X_j^n) = in\,\bar{x}_j^n, \quad l(X_j^{-n}) = i\,\frac{\partial}{\partial \bar{x}_j^n}, \quad r(X_j^n) = i\,\frac{\partial}{\partial x_j^n}, \quad r(X_j^{-n}) = in\,x_j^n.$$

These operators then satisfy the commutation relations

$$\big[l(X_j^n), l(X_k^m)\big] = n\delta_{n,-m}\delta_{j,k}, \quad \big[r(X_j^n), r(X_k^m)\big] = -n\delta_{n,-m}\delta_{j,k},$$

and $\big[l(X_j^n), r(X_k^m)\big] = 0$ for all $n, m \neq 0$. For $n = 0$, we also have left and right actions of the Lie algebra $\mathfrak{g}$ of constant loops, which acts adjointly on $S^*(W)$ and $S^*(\bar{W})$. The adjoint action on $\bar{W}^*$ and W^* is given by

$$l(X)\colon \bar{x}_j^n \mapsto \sum_k \big\langle [X, X_j], X_k \big\rangle\, \bar{x}_k^n, \quad r(X)\colon x_j^n \mapsto \sum_k \big\langle [X, X_j], X_k \big\rangle\, x_k^n.$$

In terms of anti-holomorphic and holomorphic polynomials, this becomes

$$l(X) = \sum_{n>0,j,k} \big\langle [X, X_j], X_k \big\rangle\, \bar{x}_k^n\, \frac{\partial}{\partial \bar{x}_j^n}. \quad r(X) = \sum_{n>0,j,k} \big\langle [X, X_j], X_k \big\rangle\, x_k^n\, \frac{\partial}{\partial x_j^n}.$$

We then obtain the commutators

$$\big[l(X), l(X_j^n)\big] = l([X, X_j]^n), \quad \big[r(X), r(X_j^n)\big] = r([X, X_j]^n)$$

for any $n \in \mathbb{Z}$, where we write $X^0 \equiv X$. As before, the left and right actions automatically commute with each other.

If we trivialize the bundle $\mathcal{B}$ over G in terms of a left-invariant frame, we may view sections of $\mathcal{B}$ as maps $G \to S^*(W^* \otimes \mathbb{C})$ that are G-equivariant with respect to the action $l + r$ on the fibers $S^*(W^* \otimes \mathbb{C})$. The Levi-Civita connection on $\mathcal{B}$ is then given by

$$\nabla^{\mathcal{B}}_{X_L} = \tilde{r}(X) + \frac{1}{2} \sum_{n>0,j,k} \big\langle [X, X_j], X_k \big\rangle \Big(\bar{x}_k^n\, \frac{\partial}{\partial \bar{x}_j^n} + x_k^n\, \frac{\partial}{\partial x_j^n} \Big), \tag{2.13}$$

where X_L is the left-invariant vector field taking the value X at the identity, and $\tilde{r}$ is the standard right action of $\mathfrak{g}$ on functions.

We now consider spinors in the infinite central charge case. We recall from (2.3) that the spin representation for $L\mathfrak{g}$ is $\mathcal{S} = \mathbb{S}_{\mathfrak{g}} \otimes \Lambda^*(W)$, and that the spinors we want to consider actually take values in the adjoint representation

$$\mathcal{S}^* = \mathbb{S} \otimes \Lambda^*(W^*).$$

In the limit as the level h tends to infinity, our Hilbert space of spinors (2.7) becomes

$$\begin{aligned} L^2(\mathcal{L}^\infty \otimes \mathcal{S}^*) &= L^2(G) \otimes S^*(W^* \otimes \mathbb{C}) \otimes \mathbb{S} \otimes \Lambda^*(W^*) \\ &= L^2(S \otimes \mathcal{B} \otimes \mathcal{F}), \end{aligned}$$

where $\mathcal{F}$ is a bundle over G with fibers $\Lambda^*(W^*)$, and S is the standard spin bundle over G with fibers $\mathbb{S}$. On $\mathcal{F}$, the Clifford action is given by

$$c(X_j^n) = i\,\iota(\partial/\partial x_j^n), \qquad c(X_j^{-n}) = i\,\varepsilon(dx_j^n),$$

for $n > 0$, where ι and ε denote interior and exterior multiplication respectively. These operators act trivially on the $\mathbb{S}$ component of $\mathcal{S}$. For $n = 0$, the Clifford action $c(X)$ acts normally on the $\mathbb{S}$ component and acts as $(-1)^F$ on $\mathcal{F}$. In the infinite central charge limit, the action $\widetilde{\mathrm{ad}}\, X_i^n$ of $L\mathfrak{g}_\mathbb{C}$ on $\mathcal{S}$ vanishes except for $n = 0$, in which case we have

$$\begin{aligned} \widetilde{\mathrm{ad}}\, X &= \frac{1}{4} \sum_{n \neq 0, j} c(X_j^{-n})\, c([X, X_j]^n) \\ &= \frac{1}{4} \sum_{n > 0, j, k} \big\langle (\mathrm{ad}\, X) X_j, X_k \big\rangle \big(c(X_j^{-n})\, c(X_k^n) + c(X_k^n)\, c(X_j^{-n}) \big) \\ &= -\frac{1}{2} \sum_{n > 0, j, k} \big\langle [X, X_j], X_k \big\rangle\, \varepsilon(dx_j^n)\, \iota(\partial/\partial x_k^n). \end{aligned}$$

The Levi-Civita connection on $\mathcal{F}$ is therefore given by

$$\nabla^{\mathcal{F}}_{X_L} = \tilde{r}(X) - \frac{1}{4} \sum_{n > 0, j, k} \big\langle [X, X_j], X_k \big\rangle\, \varepsilon(dx_j^n)\, \iota(\partial/\partial x_k^n).$$

Combining this with the Levi-Civita connection (2.13) on $\mathcal{B}$ and the standard spin connection ∇^S on the spin bundle S, the Levi-Civita connection on the bundle $S \otimes \mathcal{B} \otimes \mathcal{F}$ is then

$$\nabla_{X_L} = \nabla^S_{X_L} + \frac{1}{2} \sum_{j,k} \big\langle [X, X_j], X_k \big\rangle \sum_{n > 0} \Big(\bar{x}_k^n \frac{\partial}{\partial \bar{x}_j^n} + x_k^n \frac{\partial}{\partial x_j^n} - \frac{1}{2} \varepsilon(dx_k^n) \iota(\partial/\partial x_j^n) \Big).$$

We now have all the machinery we need to write down the Dirac operator for the infinite central charge case. Taking our expression (2.8) with charge $q = \frac{1}{2}$ to give the Dirac operator corresponding to the Levi-Civita connection, we obtain

$$\not{\partial} = (-1)^F \sum_j X_j \Big(\tilde{r} + r + \frac{1}{2} (\widetilde{\mathrm{ad}}_\mathbb{S} + \widetilde{\mathrm{ad}}_\mathcal{F}) \Big)(X_j) + \sum_{n \neq 0, j} c(X_j^{-n})\, r(X_j^n). \tag{2.14}$$

To obtain this expression, we took the sum $\sum_i X_i^* \nabla_{X_i}$ over a basis of left-invariant vector fields. However, the Dirac operator itself does not depend on the choice of basis. So, if we instead use a basis of right-invariant vector fields, the above discussion proceeds unchanged except for swapping the left and right actions, and we obtain the mirror image of our expression for the Dirac operator:

$$\not{\partial} = (-1)^F \sum_j X_j \Big(\tilde{l} + l + \frac{1}{2}(\widetilde{\mathrm{ad}}_{\mathbb{S}} + \widetilde{\mathrm{ad}}_{\mathcal{F}}) \Big)(X_j) + \sum_{n \neq 0, j} c(X_j^{-n})\, l(X_j^n). \tag{2.15}$$

Combining these two expressions for the Dirac operator, we take expression (2.14) for $n > 0$ and expression (2.15) for $n < 0$, and we average the two for $n = 0$. Finally, we obtain the Dirac operator ∂_0 constructed in [**7**, §3]:

$$\begin{aligned} \not{\partial} &= \sum_j c(X_j) \Big(\frac{1}{2}(\tilde{l} + \tilde{r}) + \frac{1}{2}(l + r + \widetilde{\mathrm{ad}}_{\mathbb{S}} + \widetilde{\mathrm{ad}}_{\mathcal{F}}) \Big)(X_j) \\ &\quad + \sum_{n>0, j} \big(c(X_j^{-n})\, r(X_j^n) + c(X_j^n)\, l(X_j^{-n}) \big) \\ &= \sum_j c(X_j) \nabla_{X_j}^{S \otimes \mathcal{B} \otimes \mathcal{F}} - \sum_{n>0, j} \Big(\varepsilon(dx_j^n) \frac{\partial}{\partial x_j^n} + \iota(\partial/\partial x_j^n) \frac{\partial}{\partial \bar{x}_j^n} \Big). \end{aligned}$$

In this expression, the first term is simply the Dirac operator on G twisted by the infinite dimensional vector bundle $\mathcal{B} \otimes \mathcal{F}$. The second term is the Dirac operator along the fibers of the normal bundle $\mathcal{N}G$, expressed here in terms of the Dolbeault operator as $\partial + \bar{\partial}$.

2.4. The Dirac Operator on LG/LH

Let G be a compact, connected Lie group, and let H be a closed, connected subgroup of G. We consider the space $L(G/H)$ of loops on the manifold G/H, which we view here as the homogeneous space $(LG)/(LH)$. We also take G to be simple, although our discussion generalizes to the semi-simple case with only minor modifications. To avoid global topological complications, we assume that G is simply connected, and we work at the Lie algebra level.

Let $\tilde{L}G$ be the universal central extension of LG, i.e., the central extension coming from the basic inner product on $\mathfrak{g}$. This central extension then restricts to a central extension $\tilde{L}H$ of LH, which corresponds to the restriction to $\mathfrak{h}$ of the basic inner product on $\mathfrak{g}$. Note that $\tilde{L}H$ is not the universal central extension of LH unless $H = G$. As in Section 1.3, we let $\mathfrak{p}$ be the orthogonal complement of $\mathfrak{h}$ in $\mathfrak{g}$ with respect to the basic inner product. "Looping" this decomposition, we see that $L\mathfrak{p}$ is then the orthogonal complement of $L\mathfrak{h}$ in $L\mathfrak{g}$. Letting X_i be an orthonormal basis for $\mathfrak{p}$, we take the loops $X_i^n = X_i z^n$ as an

orthonormal basis for $L\mathfrak{p}$. Viewing it as a formal element of $U(\tilde{L}\mathfrak{g}) \otimes \mathrm{Cl}(L\mathfrak{p})$, the Dirac operator on LG/LH is

$$\not{\partial} = \sum_{n,i} \big(r(X_i^n) \otimes X_i^{-n} + q \otimes X_i^{-n} \cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_i^n \big), \tag{2.16}$$

where

$$\widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X^n = \frac{1}{4} \sum_{m,j} X_j^{-m} \cdot [X, X_j]_{\mathfrak{p}}^{n+m}$$

is the spin lift of the projected adjoint representation $\mathrm{ad}_{\mathfrak{p}}\, \xi : \eta \mapsto [\xi, \eta]_{L\mathfrak{p}}$.

We now consider the diagonal representation $\varrho : \tilde{L}\mathfrak{h} \to U(\tilde{L}\mathfrak{g}) \otimes \mathrm{Cl}(L\mathfrak{p})$ given by $\varrho = r \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}$, where

$$\widetilde{\mathrm{ad}}\, Z^n = \frac{1}{4} \sum_{m,j} X_j^{-m} \cdot [Z, X_j]^{n+m}$$

for any $Z^n \in L\mathfrak{h}$. This representation corresponds to the right action of $\tilde{L}\mathfrak{h}$ on spinors, and in order that our Dirac operator descend to an operator on LG/LH, we require that it commutes with this action. We recall from Section 2.2 that the Dirac operator on the loop group LG does not actually commute with the corresponding diagonal action. This is because the Dirac operator we constructed on LG is actually the Dirac operator for the homogeneous space $\tilde{L}G/S^1$, and it failed to commute due to additional corrections coming from the central extension. For the Dirac operator on LG/LH, if we take into account the central extensions we have $\tilde{L}G/\tilde{L}H \cong LG/LH$, and so the corrections are already accounted for in the $\tilde{L}\mathfrak{h}$-action.

Lemma 5. *The Dirac operator $\not{\partial}$ on LG/LH commutes with the elements*

$$\varrho(\zeta) = r(\zeta) \otimes 1 + 1 \otimes \widetilde{\mathrm{ad}}\, \zeta \in U(\tilde{L}\mathfrak{g}) \otimes \mathrm{Cl}(L\mathfrak{p})$$

for all $\zeta \in \mathbb{R} \tilde{\oplus} \tilde{L}\mathfrak{h}$.

Proof. First we note that $\not{\partial}$ automatically commutes with the central element $\varrho(I)$. As we saw in (2.9), the energy operator $\varrho(\partial_\theta)$ commutes with the Dirac operator on LG, and the same also holds true in the homogeneous case for the Dirac operator on LG/LH. For the elements $\varrho(Z^n)$, we proceed here exactly as we did for the proofs of Lemmas 3 and 4 in the compact Lie

group case. We first compute the commutator

$$\begin{aligned}\Big[\varrho(Z^n), \sum_{m,j} r(X_j^m)\otimes X_j^{-m}\Big] &= \sum_{m,j}\Big([r(Z^n), r(X_j^m)]\otimes X_j^{-m} + r(X_j^m)\otimes[\widetilde{\mathrm{ad}}\, Z^n, X_j^{-m}]\Big)\\ &= \sum_{m,j} r([Z,X_j]^{n+m})\otimes X_j^{-m} + \sum_{m,j} r(X_j^m)\otimes [Z,X_j]^{n-m} = 0.\end{aligned}$$

Note that there is no central extension term $r(I)$ here since $\langle Z, X\rangle = 0$ for all $Z\in\mathfrak{h}$ and $X\in\mathfrak{p}$. For the cubic term of the Dirac operator we have

$$\begin{aligned}\Big[\widetilde{\mathrm{ad}}\, Z^n, \sum_{m,j} X_j^{-m}\cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_j^m\Big] &= \sum_{m,j}\Big([\widetilde{\mathrm{ad}}\, Z^n, X_j^{-m}]\cdot \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_j^m + X_j^{-m}\cdot[\widetilde{\mathrm{ad}}\, Z^n, \widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_j^m]\Big)\\ &= \sum_{m,j}[Z,X_j]^{n-m}\cdot\widetilde{\mathrm{ad}}_{\mathfrak{p}}\, X_j^m + \sum_{m,j} X_j^{-m}\cdot\widetilde{\mathrm{ad}}_{\mathfrak{p}}[Z,X_j]^{n+m} = 0.\end{aligned}$$

Combining these two commutators, we obtain the result $[\varrho(Z^n), \not{\partial}] \equiv 0$. □

We now want to interpret our Dirac operator as an operator acting on spinors on LG/LH. In order to do this, we must first examine the spin representation $\mathcal{S}_{L\mathfrak{p}}$ corresponding to $L\mathfrak{p}$. Since we have the orthogonal decomposition $L\mathfrak{g} = L\mathfrak{h}\oplus L\mathfrak{p}$, the spin representation $\mathcal{S}_{L\mathfrak{g}}$ corresponding to $L\mathfrak{g}$ decomposes as the tensor product

$$\mathcal{S}_{L\mathfrak{g}} = \mathcal{S}_{L\mathfrak{h}}\otimes\mathcal{S}_{L\mathfrak{p}},$$

provided that $\dim\mathfrak{p}$ is even (which automatically holds if $\mathfrak{h}$ is a maximal rank subalgebra of $\mathfrak{g}$). Recalling our description (2.3) of $\mathcal{S}_{L\mathfrak{g}}$, we obtain

$$\mathcal{S}_{L\mathfrak{p}} = \mathbb{S}_{\mathfrak{p}}\oplus\Lambda^*\big(\textstyle\bigoplus_{n>0}\mathfrak{p}_{\mathbb{C}}z^n\big). \tag{2.17}$$

The level of $\mathcal{S}_{L\mathfrak{p}}$ is then $c_G - c_H$, where c_H here denotes the level of $\mathcal{S}_{L\mathfrak{h}}$ with regard to the central extension $\tilde{L}H$. From here on, we write $\mathcal{S}$ for $\mathcal{S}_{L\mathfrak{p}}$.

Remark. We recall that $\tilde{L}H$ is not the universal central extension of LH. If H is semi-simple, the universal central extension $\hat{L}H$ is actually an extension of LH by a torus $(S^1)^k$ containing one copy of S^1 for each of the simple components of H. The spin representation $\mathcal{S}_{L\mathfrak{h}}$ is then a positive energy representation of $\hat{L}H$ of level $(c_{H_1},\dots,c_{H_k})$, where $H_1,\dots,H_k$ are the simple components of H. However, this representation of $\hat{L}H$ does not necessarily descend to a representation of the central extension $\tilde{L}H$. There are topological obstructions, corresponding to the characteristic classes w_2 and $\frac{1}{2}p_1$ of G/H,

and we obtain representations of $\tilde{L}H$ on $\mathcal{S}_{L\mathfrak{h}}$ and $\mathcal{S}_{L\mathfrak{p}}$ if and only if G/H admits a string structure, the loop space analogue of a spin structure. For a discussion of string structures, see [**5**]. To consider spinors on the loop space of a general manifold, Witten suggests in [**9**] twisting with a vector bundle that cancels these obstructions, giving the loop space analogue of a Spin^c structure.

For the Hilbert space of spinors on LG/LH, we first consider the space

$$L^2(\mathcal{L}^{h+c_G-c_H} \otimes \mathcal{S}^*) = \widehat{\bigoplus}_\lambda \mathcal{H}_{\lambda,h} \otimes \left(\mathcal{H}^*_{\lambda,h} \otimes \mathcal{S}^*\right)^{L\mathfrak{h}}$$

of sections invariant under the diagonal action ϱ of $L\mathfrak{h}$. However, the product representation $\mathcal{H}^*_{m,\lambda,h} \otimes \mathcal{S}^*$ has level $-h - c_G + c_H$, which is strictly less than zero. Since the trivial representation of $L\mathfrak{h}$ has level zero, there cannot be any $L\mathfrak{h}$-invariant sections, and thus this Hilbert space is trivial!

To obtain a non-trivial Hilbert space of spinors, we instead consider spinors twisted by a positive energy representation $\mathcal{U}$ of $L\mathfrak{h}$ of level $k \geq c_G - c_H$. The Hilbert space of twisted spinors is then

$$\begin{aligned}\left(L^2(\tilde{L}G) \otimes \mathcal{S}^* \otimes \mathcal{U}\right)^{L\mathfrak{h}} &= \widehat{\bigoplus}_\lambda \mathcal{H}_{\lambda,h} \otimes \left(\mathcal{H}^*_{\lambda,h} \otimes \mathcal{S}^* \otimes \mathcal{U}\right)^{L\mathfrak{h}} \\ &= \widehat{\bigoplus}_\lambda \mathcal{H}_{\lambda,h} \otimes \mathrm{Hom}_{L\mathfrak{h}}\left(\mathcal{H}_{\lambda,h} \otimes \mathcal{S}, \mathcal{U}\right),\end{aligned}$$

where $h = k - c_G + c_H \geq 0$. The space $\mathrm{Hom}_{L\mathfrak{h}}\left(\mathcal{H}_{\lambda,h} \otimes \mathcal{S}, \mathcal{U}\right)$ of $L\mathfrak{h}$-equivariant homomorphisms is infinite dimensional, but we can further decompose it into its constant energy subspaces with respect to the action of ∂_θ, the generator of rotations of the circle. Letting $\mathbb{C}_l$ be the one dimensional representation where ∂_θ acts by il, we have

$$\begin{aligned}\left(L^2(\tilde{L}G) \otimes \mathcal{S}^* \otimes \mathcal{U}\right)^{L\mathfrak{h}} &= \widehat{\bigoplus}_{l,\lambda} \mathcal{H}_{\lambda,h} \otimes \mathrm{Hom}_{L\mathfrak{h}}\left(\mathcal{H}_{\lambda,h} \otimes \mathcal{S}, \mathcal{U}\right)(l) \\ &= \widehat{\bigoplus}_{l,\lambda} \mathcal{H}_{\lambda,h} \otimes \mathbb{C}_l \otimes \mathrm{Hom}_{\mathbb{R}\,\tilde{\oplus}\,L\mathfrak{h}}\left(\mathcal{H}_{\lambda,h} \otimes \mathbb{C}_l \otimes \mathcal{S}, \mathcal{U}\right) \\ &= \widehat{\bigoplus}_{l,\lambda} \mathcal{H}_{l,\lambda,h} \otimes \mathrm{Hom}_{\mathbb{R}\,\tilde{\oplus}\,L\mathfrak{h}}\left(\mathcal{H}_{l,\lambda,h} \otimes \mathcal{S}, \mathcal{U}\right),\end{aligned}$$

where $\mathcal{H}_{l,\lambda,h} \cong \mathcal{H}_{\lambda,h} \otimes \mathbb{C}_l$ is the positive energy representation with lowest weight (l, λ, h). In other words, if we consider sections invariant under the action of the semi-direct sum $\mathbb{R}\,\tilde{\oplus}\,L\mathfrak{h}$, then we must sum over all positive energy representations with the full spectrum of minimum energies, instead of first normalizing them to have minimum energy zero.

Since both $\mathcal{H}_{l,\lambda,h}$ and $\mathcal{S}$ are positive energy representations of finite type (i.e., each of their constant energy subspaces is finite dimensional), their tensor product $\mathcal{H}_{l,\lambda,h} \otimes \mathcal{S}$ is again a positive energy representation of finite type. This

tensor product is therefore completely reducible into an infinite direct sum of irreducible positive energy representations, with only finitely many terms at each minimum energy. In particular, if $\mathcal{U}$ is an irreducible positive energy representation of $L\mathfrak{h}$, then each of the vector spaces $\mathrm{Hom}_{\mathbb{R}\,\tilde{\oplus}\,L\mathfrak{h}}(\mathcal{H}_{l,\lambda,h}\otimes\mathcal{S},\mathcal{U})$ is finite dimensional.

As always, the spin representation $\mathcal{S}$ decomposes into the direct sum of the two half-spin representations $\mathcal{S}\cong\mathcal{S}^+\oplus\mathcal{S}^-$. With respect to our expression (2.17) for $\mathcal{S}$, we have

$$\mathcal{S}^+\cong\big(\mathbb{S}^+_{\mathfrak{p}}\otimes\Lambda^{\mathrm{even}}\big)\oplus\big(\mathbb{S}^-_{\mathfrak{p}}\otimes\Lambda^{\mathrm{odd}}\big),$$
$$\mathcal{S}^-\cong\big(\mathbb{S}^-_{\mathfrak{p}}\otimes\Lambda^{\mathrm{even}}\big)\oplus\big(\mathbb{S}^+_{\mathfrak{p}}\otimes\Lambda^{\mathrm{odd}}\big).$$

If $\dim\mathfrak{p}$ is even, then the Clifford action of $L\mathfrak{p}$ on $\mathcal{S}$ interchanges $\mathcal{S}^+$ and $\mathcal{S}^-$. Let $\mathcal{U}_{m,\mu,k}$ be the irreducible positive energy representation of $L\mathfrak{h}$ with lowest weight (m,μ,k), with level $k\geq c_G-c_H$ and minimum energy m. (Since we are now considering representations of $L\mathfrak{g}$ with a full spectrum of minimum energies, we do the same for representations of $L\mathfrak{h}$.) We define the Dirac operator on LG/LH twisted by the representation $\mathcal{U}_{m,\mu,k}$ to be the operator

$$\not{\partial}^*\otimes\mathcal{U}_{m,\mu,k}:\big(L^2(\tilde{L}G)\otimes(\mathcal{S}^+)^*\otimes\mathcal{U}_{m,\mu,k}\big)^{L\mathfrak{h}}\to\big(L^2(\tilde{L}G)\otimes(\mathcal{S}^-)^*\otimes\mathcal{U}_{m,\mu,k}\big)^{L\mathfrak{h}}$$

given by (2.16), where the right action r acts on the $\mathcal{H}^*_{l,\lambda,h}$ components in the decomposition (2.5) of $L^2(\tilde{L}G)$, the Clifford algebra $\mathrm{Cl}(L\mathfrak{p})$ acts via the Clifford action on $(\mathcal{S}^\pm)^*$, and the Dirac operator acts trivially on the $\mathcal{U}_{m,\mu,k}$ component.

2.5. Induced Representations of Loop Groups

In order to make sense of the representations we used to define the domain of the Dirac operator in the previous section, we introduce some new machinery, giving the loop group analogue of the theory developed in [**2**] for compact Lie groups. First we require a few definitions.

Definition. The *representation ring* $R(LG)$ of a loop group LG is the ring of isomorphism classes of virtual positive energy representations of LG of finite type.

In terms of the irreducible positive energy representations, the elements of $R(LG)$ can be written as possibly infinite sums $\sum_i a_i\,\mathcal{H}_{m_i,\lambda_i,h_i}$ with coefficients $a_i\in\mathbb{Z}$. The positive energy condition requires that the minimum energies m_i of the summands be bounded below, and the finite type condition requires that there be only finitely many terms for each minimum energy. In other words, if the sum is infinite, then the energies m_i tend to infinity. The ring structure on $R(LG)$ is induced by the direct sum and tensor product for positive energy

representations. The level h of the positive energy representations then gives us a grading $R(LG) = \bigoplus_{h\geq 0} R_h(LG)$ of the representation ring.

Definition. Let $\hat{R}(LG)$ be the group of possibly infinite formal sums $\sum_i a_i \mathcal{H}_{m_i,\lambda_i,h_i}$ with coefficients $a_i \in \mathbb{Z}$, where the minimum energies m_i are bounded above and there are only finitely many terms for each minimum energy.

In other words, for an infinite element of $\hat{R}(LG)$, the minimum energies of the summands tend to negative infinity. Like the representation ring, $\hat{R}(LG)$ is a group with repect to addition, but we no longer have a multiplication map since the coefficients of a product could be infinite. Again, we have a decomposition $\hat{R}(LG) = \bigoplus_{h\geq 0} \hat{R}_h(LG)$ according to the level h. We also note that the intersection $R(LG) \cap \hat{R}(LG)$, consisting of all finite linear combinations, is the free abelian group generated by the irreducible positive energy representations. For these finite elements, we can define an inner product modeled on the intertwining number of two positive energy representations

$$\langle \mathcal{H}_1, \mathcal{H}_2 \rangle = \dim_{\mathbb{C}} \operatorname{Hom}_{S^1 \ltimes LG}(\mathcal{H}_1, \mathcal{H}_2),$$

with respect to which the irreducible positive energy representations correspond to an orthonormal basis. This inner product extends to a duality pairing of $R(LG)$ with $\hat{R}(LG)$ since any two elements of $R(LG)$ and $\hat{R}(LG)$ can have only finitely many summands in common.

Given any two positive energy representations, their tensor product is again a positive energy representation, and the same is true for negative energy representations. However, if we take the tensor product of a positive energy representation with a negative energy representation, the resulting representation contains the full spectrum of energies, and so it is neither a positive nor a negative energy representation. However, it does contain subspaces that are built entirely out of either positive or negative energy representations.

Example (Second Quantization). Let $G = S^1$, and let $\mathcal{H}_h$ be the irreducible positive energy representation of $\tilde{L}S^1$ of level h. Taking the basis $\{H^n = Hz^n\}_{n\in\mathbb{Z}}$ for the complexified Lie algebra of LS^1, the action ρ_h on $\mathcal{H}_h$ satisfies the commutation relations

$$[\rho_h(H^n), \rho_h(H^m)] = nh\, \delta_{n,-m}.$$

(From here on, we omit ρ_h.) As we saw in Section 2.3, we can identify $\mathcal{H}_h$ with the space of polynomials in the variables $z_1, z_2, z_3, \ldots$, where H^n acts by multiplication by $ihnz_n$ and H^{-n} acts by $i\frac{\partial}{\partial z_n}$.

Consider now the tensor product $\mathcal{H}_h \otimes \mathcal{H}^*_{-h}$. Here $\mathcal{H}_{-h}$ is a positive energy representation of level $-h$ (since LS^1 is abelian, we do not need to require that the level be non-negative), and its dual $\mathcal{H}^*_{-h}$ is a negative energy representation

of level h. We can identify this tensor product with the space of polynomials in both the z_i and their conjugates $\bar{z}_i$. Here, the diagonal Lie algebra action is of level $2h$, and it is given by

$$H^n \mapsto i\left(\frac{\partial}{\partial \bar{z}_n} + hnz_n\right), \qquad H^{-n} \mapsto i\left(\frac{\partial}{\partial z_n} - hn\bar{z}_n\right).$$

In order to write this product in terms of positive energy representations, consider the power series

$$\Phi = \exp\left(h\sum_{n>0} n\,|z_n|^2\right).$$

We then have $H^{-n}\Phi = 0$ for all $n > 0$, so Φ is a lowest weight vector generating an irreducible positive energy representation of level $2h$ living inside $\mathcal{H}_h \otimes \mathcal{H}^*_{-h}$. Furthermore, this tensor product admits a second $\tilde{L}S^1$-action, commuting with the first, given by

$$H^n \mapsto i\left(\frac{\partial}{\partial \bar{z}_n} - hnz_n\right), \qquad H^{-n} \mapsto i\left(\frac{\partial}{\partial z_n} + hn\bar{z}_n\right).$$

For this action, we see that H^n annihilates Φ, so Φ now generates an irreducible negative energy representation of level $-2h$. Combining these actions, we can express the tensor product $\mathcal{H}_h \otimes \mathcal{H}^*_{-h}$ as the representation $\mathcal{H}_{2h} \otimes \mathcal{H}^*_{2h}$ of $\tilde{L}S^1 \times \tilde{L}S^1$. In other words, this product decomposes as an infinite sum of positive energy representations where the minimum energy vectors of the summands themselves lie in a negative energy representation, which is precisely the condition for it to give an element of $\hat{R}(LS^1)$. In string theory, this is second quantization on the circle, and the two $\tilde{L}S^1$-actions correspond to the left and right moving bosons.

In general, given an arbitrary representation $\mathcal{H}$ of LG which is not necessarily positive or negative energy, we let $\mathcal{H}_0$ denote the subspace of all lowest weight vectors in $\mathcal{H}$. In other words, $\mathcal{H}_0$ is the subspace of $\mathcal{H}$ annihilated by the root spaces $\mathfrak{g}_\alpha z^n \subset L\mathfrak{g}_\mathbb{C}$ for all negative roots $(n, \alpha) \in \Phi^-(LG)$. We then define the positive energy part $\mathcal{H}_+$ to be the subspace of $\mathcal{H}$ generated by $\mathcal{H}_0$ with respect to the LG-action on $\mathcal{H}$. More precisely, we have $\mathcal{H}_+ = U(\tilde{L}\mathfrak{g})\mathcal{H}_0$. Now, if v is a lowest weight vector in $\mathcal{H}_0$ with weight (m, λ, h), then v generates a subspace isomorphic to the positive energy representation $\mathcal{H}_{m,\lambda,h}$. Decomposing $\mathcal{H}_0$ into its weight spaces for the action of $S^1 \times T \times S^1$, we can therefore write the positive energy part of $\mathcal{H}$ as

$$\mathcal{H}_+ = \bigoplus_{m,\lambda,h} \mathcal{H}_{m,\lambda,h} \otimes \mathrm{Hom}_{S^1 \ltimes LG}(\mathcal{H}_{m,\lambda,h}, \mathcal{H}).$$

If $\mathcal{H}$ is a positive energy representation to begin with, then we have $\mathcal{H}_+ = \mathcal{H}$, and this expression is just the decomposition of $\mathcal{H}$ into a direct sum of irreducible positive energy representations. On the other hand, if $\mathcal{H}$ is a negative energy representation, then $\mathcal{H}_+ = 0$.

LEMMA 6. *The map taking a pair of positive energy representations $\mathcal{H}_1, \mathcal{H}_2$ to $(\mathcal{H}_1 \otimes \mathcal{H}_2^*)_+$ induces a bilinear map $\hat{R}(LG) \times R(LG) \to \hat{R}(LG)$.*

PROOF. We first consider the case where both $\mathcal{H}_1$ and $\mathcal{H}_2$ are irreducible. We then have

$$\mathrm{Hom}_{S^1 \ltimes LG}(\mathcal{H}_{m,\lambda,h}, \mathcal{H}_1 \otimes \mathcal{H}_2^*) \cong \mathrm{Hom}_{S^1 \ltimes LG}(\mathcal{H}_{m,\lambda,h} \otimes \mathcal{H}_2, \mathcal{H}_1). \tag{2.18}$$

The tensor product $\mathcal{H}_{m,\lambda,h} \otimes \mathcal{H}_2$ is a positive energy representation of finite type, which is completely reducible as a direct sum of irreducible positive energy representations, with only finitely many summands for each minimum energy. Therefore, the vector space given by (2.18) is finite dimensional. Furthermore, letting m_1 and m_2 be the minimum energies of $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively, we see that the minimum energy of $\mathcal{H}_{m,\lambda,h} \otimes \mathcal{H}_2$ is $m + m_2$, and thus the vector space is trivial for $m_1 < m + m_2$.

Working in the representation ring, the element $(\mathcal{H}_1 \otimes \mathcal{H}_2^*)_+$ becomes

$$(\mathcal{H}_1 \otimes \mathcal{H}_2^*)_+ = \sum_{m,\lambda,h} \mathcal{H}_{m,\lambda,h} \langle \mathcal{H}_{m,\lambda,h} \mathcal{H}_2, \mathcal{H}_1 \rangle, \tag{2.19}$$

which is a well-defined formal sum for any $\mathcal{H}_1 \in \hat{R}(LG)$ and $\mathcal{H}_2 \in R(LG)$. Now letting m_1 and m_2 be the highest and lowest minimum energies in $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively, we see that the coefficient of $\mathcal{H}_{m,\lambda,h}$ vanishes for $m > m_1 - m_2$, and so the minimum energies of the summands are bounded above. Finally, taking into account the grading on $R(LG)$ and $\hat{R}(LG)$ according to the level h, and recalling that there are only finitely many irreducible positive energy representations at each level with a given minimum energy, we see that there are only finitely many terms in the sum (2.19) for each value of m. It then follows that the formal sum $(\mathcal{H}_1 \otimes \mathcal{H}_2^*)_+$ gives an element of $\hat{R}(LG)$. □

Let H be a Lie subgroup of G. We have an inclusion $i : LH \to LG$, and any action of LG automatically restricts to an action of LH. This restriction map gives a ring homomorphism $i^* : R(LG) \to R(LH)$, and we can consider its formal adjoint with respect to the pairing $R(LG) \times \hat{R}(LG) \to \mathbb{Z}$.

DEFINITION. The *induction map* $i_* : \hat{R}(LH) \to \hat{R}(LG)$ is given by

$$i_* : \mathcal{U} \mapsto \sum_{m,\lambda,h} \langle \mathcal{U}, i^* \mathcal{H}_{m,\lambda,h} \rangle_{LH} \mathcal{H}_{(m,\lambda,h)}, \tag{2.20}$$

where we sum over all irreducible positive energy representations of LG.

Given any representation $\mathcal{M}$ of LH, we can consider the induced bundle $LG \times_{LH} \mathcal{M}$ over the homogeneous space LG/LH. For the Hilbert space of sections of this bundle, we take

$$\left(L^2(\tilde{L}G) \otimes \mathcal{M}\right)^{LH} \cong \widehat{\bigoplus}_{m,\lambda,h} \mathcal{H}_{m,\lambda,h} \otimes \mathrm{Hom}_{S^1 \ltimes LH}\left(\mathcal{H}_{m,\lambda,h}, \mathcal{M}\right), \tag{2.21}$$

where $L^2(\tilde{L}G)$ is defined by (2.5). The left action of LG on $L^2(\tilde{L}G)$ persists as an action on these sections, corresponding to the action of LG on the $\mathcal{H}_{m,\lambda,h}$ components on the right side of (2.21). Also, we start out with a right action of the semi-direct product $S^1 \ltimes LH$ on the product $L^2(\tilde{L}G) \otimes \mathcal{M}$, and when we restrict to the LH-invariant subspace, we are still left with a residual S^1-action. This is manifested on the right side of (2.21) as the shifted minimum energy m of the positive energy representation $\mathcal{H}_{m,\lambda,h}$. Comparing equations (2.20) and (2.21), we see that if $\mathcal{M}_+$ corresponds to an element of $\hat{R}(LH)$, then the Hilbert space of sections is the induced representation of LG corresponding to the element $i_*(\mathcal{M}_+) \in \hat{R}(LG)$.

THEOREM 5. *Let $\mathcal{M}$, $\mathcal{N}$ be LH-representations with positive energy parts $\mathcal{M}_+$, $\mathcal{N}_+$ corresponding to elements of $\hat{R}(LG)$, and let D be a linear operator*

$$D : \left(L^2(\tilde{L}G) \otimes \mathcal{M}\right)^{LH} \to \left(L^2(\tilde{L}G) \otimes \mathcal{N}\right)^{LH}$$

commuting with both the left LG-action and the right S^1-action on the domain and range. The LG-equivariant index of D is the virtual LG representation

$$\mathrm{Index}_{LG}(D) = i_*(\mathcal{M}_+ - \mathcal{N}_+) \in \hat{R}(LG),$$

corresponding to the difference between the domain and the range of D.

PROOF. Since the operator D commutes with the action of LG, both its kernel and cokernel are invariant subspaces for the LG-action. Therefore, the index of D is a virtual representation of LG. Also, since D commutes with the actions of LG and S^1, it respects the decomposition of the domain and range given by (2.21). In particular, we can write D as an infinite sum of operators

$$D_{\boldsymbol{\lambda}} : \mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}) \to \mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{N}),$$

indexed by the anti-dominant weights $\boldsymbol{\lambda} = (m, \lambda, h)$ of LG. We can further decompose the positive energy representations $\mathcal{H}_{\boldsymbol{\lambda}}$ into their constant energy eigenspaces $\mathcal{H}_{\boldsymbol{\lambda}}(k)$, each of which is a finite dimensional representation of G (as the irreducible positive energy representations are of finite type). Since we require that $\mathcal{M}_+$ and $\mathcal{N}_+$ both correspond to elements of $\hat{R}(LH)$, the vector spaces $\mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M})$ and $\mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{N})$ are finite dimensional as well. The operators $D_{\boldsymbol{\lambda}}$ can thus be broken down into an infinite family of

G-equivariant finite dimensional operators

$$D_{\boldsymbol{\lambda},k} : \mathcal{H}_{\boldsymbol{\lambda}}(k) \otimes \mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}) \to \mathcal{H}_{\boldsymbol{\lambda}}(k) \otimes \mathrm{Hom}_{S^1 \ltimes LH}(\mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{N}).$$

Now, for a finite dimensional operator, the index is simply the difference in dimension between the domain and the range, and for a G-equivariant operator, the G-index is the difference between the domain and the range viewed as a virtual representation of G. For each of the operators $D_{\boldsymbol{\lambda},k}$ we have

$$\mathrm{Index}_{S^1 \times G}(D_{\boldsymbol{\lambda},k}) = \mathcal{H}_{\boldsymbol{\lambda}}(k) \, \langle i^* \mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}_+ - \mathcal{N}_+ \rangle_{LH}.$$

The index of the operator $D_{\boldsymbol{\lambda}}$ is then the sum of the indices of its components $D_{\boldsymbol{\lambda},k}$, and as a virtual $S^1 \times G$ representation, we obtain

$$\mathrm{Index}_{S^1 \times G}(D_{\boldsymbol{\lambda}}) = \mathcal{H}_{\boldsymbol{\lambda}} \, \langle i^* \mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}_+ - \mathcal{N}_+ \rangle_{LH}.$$

However, each of the operators $D_{\boldsymbol{\lambda}}$ is LG-equivariant, and so its index is actually a representation of LG. The only possible LG-representation is then

$$\mathrm{Index}_{LG}(D_{\boldsymbol{\lambda}}) = \mathcal{H}_{\boldsymbol{\lambda}} \, \langle i^* \mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}_+ - \mathcal{N}_+ \rangle_{LH}.$$

Finally, the index of the original operator D is the sum of the indices of each of its components $D_{\boldsymbol{\lambda}}$, which gives us

$$\mathrm{Index}_{LG}(D) = \sum_{\boldsymbol{\lambda}} \mathcal{H}_{\boldsymbol{\lambda}} \, \langle i^* \mathcal{H}_{\boldsymbol{\lambda}}, \mathcal{M}_+ - \mathcal{N}_+ \rangle_{LH} = i_*(\mathcal{M}_+ - \mathcal{N}_+).$$

which is just the virtual representation of LG corresponding to the difference of the domain and range of the operator D. □

Now that we can compute the index of a homogeneous operator on LG/LH, we apply this result to the Dirac operator. Recall from the previous section that the Dirac operator twisted by a representation $\mathcal{U}$ of LH is an operator

$$\not{\partial}^* \otimes \mathcal{U} : \left(L^2(\tilde{L}G) \otimes (\mathcal{S}^+)^* \otimes \mathcal{U}\right)^{L\mathfrak{h}} \to \left(L^2(\tilde{L}G) \otimes (\mathcal{S}^-)^* \otimes \mathcal{U}\right)^{L\mathfrak{h}}.$$

If $\mathcal{U}$ corresponds to an element of $\hat{R}(L\mathfrak{h})$, then the positive energy part of the tensor product $(\mathcal{U} \otimes (\mathcal{S}^\pm)^*)_+$ gives an element of $\hat{R}(L\mathfrak{h})$. In particular, we will consider the case where $\mathcal{U}$ is an irreducible positive energy representation of $L\mathfrak{h}$. By Theorem 5, the $L\mathfrak{h}$-equivariant index of the Dirac operator is then

$$\begin{aligned}
\mathrm{Index}_{L\mathfrak{g}} \not{\partial}^* \otimes \mathcal{U} &= i_*(\mathcal{U}(\mathcal{S}^+)^* - \mathcal{U}(\mathcal{S}^-)^*)_+ \\
&= \sum\nolimits_{\boldsymbol{\lambda}} \mathcal{H}_{\boldsymbol{\lambda}} \left\langle i^*(\mathcal{H}_{\boldsymbol{\lambda}}), \mathcal{U}\,(\mathcal{S}^+ - \mathcal{S}^-)^* \right\rangle_{L\mathfrak{h}} \\
&= \sum\nolimits_{\boldsymbol{\lambda}} \mathcal{H}_{\boldsymbol{\lambda}} \left\langle i^*(\mathcal{H}_{\boldsymbol{\lambda}})(\mathcal{S}^+ - \mathcal{S}^-), \mathcal{U} \right\rangle_{L\mathfrak{h}}.
\end{aligned}$$

So, in order to explictly compute the index of the twisted Dirac operator, we must first decompose the virtual $L\mathfrak{h}$-representation

$$\mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathcal{S}^+ - \mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathcal{S}^-$$

as a linear combination of irreducible positive energy representations of $L\mathfrak{h}$. This is the purpose of the following section.

2.6. The Homogeneous Weyl-Kac Formula

We begin by reviewing the Weyl character formula. Given a vector space V, let $\mathbb{S}_V$ be the spin representation of $\mathrm{Spin}(V)$, and denote the corresponding half-spin representations by $\mathbb{S}_V^{\pm}$. Let G be a compact semi-simple Lie group with maximal torus T. Given an Ad-invariant inner product on $\mathfrak{g}$, we abuse notation and write $\mathfrak{g}/\mathfrak{t}$ for the orthogonal complement of $\mathfrak{t}$ in $\mathfrak{g}$. Then $\mathfrak{g}/\mathfrak{t}$ is a T-representation, and the T-action lifts to an action on $\mathbb{S}_{\mathfrak{g}/\mathfrak{t}}$. The Weyl character formula states that if V_λ is the irreducible representation of G with highest weight λ, then its character satisfies the identity

$$\chi(V_\lambda)\,\chi(\mathbb{S}^+_{\mathfrak{g}/\mathfrak{t}} - \mathbb{S}^-_{\mathfrak{g}/\mathfrak{t}}) = \sum_{w\in W_G} (-1)^w e^{w(\lambda+\rho_G)},$$

where ρ_G is half the sum of the positive roots of G and is also the highest weight of $\mathbb{S}_{\mathfrak{g}/\mathfrak{t}}$. In particular, we have the identity

$$\chi(\mathbb{S}^+_{\mathfrak{g}/\mathfrak{t}} - \mathbb{S}^-_{\mathfrak{g}/\mathfrak{t}}) = \sum_{w\in W_G} (-1)^w e^{w(\rho_G)}.$$

For loop groups, the Weyl-Kac character formula is completely analogous. Letting G and T be as before, we now consider the representation $\mathcal{H}_{m,\lambda,h}$ of LG with lowest weight (m,λ,h). Recall that we take $S^1\times T\times S^1$ for the maximal torus of $S^1\ltimes \tilde{L}G$, where the first S^1 corresponds to rotation of loops and the second S^1 to the central extension. The Lie algebra of LG is $L\mathfrak{g}$, and we write $L\mathfrak{g}/\mathfrak{t}$ for the orthogonal complement of $\mathfrak{t}$ in $L\mathfrak{g}$. The corresponding spin space $\mathcal{S}_{L\mathfrak{g}/\mathfrak{t}}$ is now infinite dimensional, but as a representation of $S^1\times T\times S^1$ it can be written in the form

$$\mathcal{S}_{L\mathfrak{g}/\mathfrak{t}} = u^{c_G} z^{m_G}\,\mathbb{S}_{\mathfrak{g}/\mathfrak{t}} \otimes \bigotimes_{n>0} \Lambda^*(\mathfrak{g}_{\mathbb{C}} z^n),$$

where z keeps track of the energy, and u keeps track of the level. We recall from Theorem 4 that the level of the spin representation is $c_G = \langle \alpha_{\max}, \rho_G\rangle + 1$ (with respect to the basic inner product on $\mathfrak{g}$) and that its minimum energy is $m_G = \frac{1}{24}\dim\mathfrak{g}$.

The half-spin representations $\mathcal{S}^{\pm}_{L\mathfrak{g}/\mathfrak{t}}$ make sense in this context, and the character of their difference can be computed from the above expression for $\mathcal{S}_{L\mathfrak{g}/\mathfrak{t}}$. The Weyl-Kac character formula then asserts that

$$\chi(\mathcal{H}_{m,\lambda,h})\,\chi(\mathcal{S}^+_{L\mathfrak{g}/\mathfrak{t}} - \mathcal{S}^-_{L\mathfrak{g}/\mathfrak{t}}) = \sum_{w\in \mathcal{W}_G} (-1)^w e^{w(m+m_G,\lambda-\rho_G,h+c_G)},$$

where the sum is now taken over the affine Weyl group $\mathcal{W}_G$ of G.

Let $\mathfrak{h}$ be a maximal rank reductive subalgebra of $\mathfrak{g}$. As we discussed at the end of Section 1.4, Gross, Kostant, Ramond, and Sternberg associate in [**3**] to each $\mathfrak{g}$-representation a set of $\mathfrak{h}$-representations with similar properties, called a multiplet. These results have immediate analogues for loop groups. Let $D_{\mathfrak{g}}$ and $D_{\mathfrak{h}}$ be the fundamental Weyl alcoves for $\mathfrak{g}$ and $\mathfrak{h}$, and consider the subset $\mathcal{C} \subset \mathcal{W}_G$, containing those elements $c \in \mathcal{W}_{\mathfrak{g}}$ that map $D_{\mathfrak{g}}$ into $D_{\mathfrak{h}}$. The cardinality of C is then the ratio of the volumes of $D_{\mathfrak{g}}$ and $D_{\mathfrak{h}}$, which is finite if and only if $\mathfrak{h}$ is itself semi-simple.

THEOREM 6. *Let $\mathcal{H}_{\boldsymbol{\lambda}}$ be the irreducible positive energy representation of $L\mathfrak{g}$ with lowest weight $\boldsymbol{\lambda} = (m, \lambda, h)$. Working with virtual representations of $L\mathfrak{h}$, we have the identity*

$$\mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathcal{S}^+_{L\mathfrak{g}/L\mathfrak{h}} - \mathcal{H}_{\boldsymbol{\lambda}} \otimes \mathcal{S}^-_{L\mathfrak{g}/L\mathfrak{h}} = \sum_{c \in \mathcal{C}} (-1)^c \mathcal{U}_{c(\boldsymbol{\lambda}+\boldsymbol{\rho}_G)-\boldsymbol{\rho}_H}, \tag{2.22}$$

where $\boldsymbol{\rho}_{\mathfrak{g}} = (m_{\mathfrak{g}}, -\rho_{\mathfrak{g}}, c_{\mathfrak{g}})$ and $\boldsymbol{\rho}_{\mathfrak{h}} = (m_{\mathfrak{h}}, -\rho_{\mathfrak{h}}, c_{\mathfrak{h}})$, and we write $\mathcal{U}_{\boldsymbol{\mu}}$ for the irreducible positive energy representation of $L\mathfrak{h}$ with lowest weight $\boldsymbol{\mu}$.

PROOF. We first note that the construction of the spin space is multiplicative (provided that the underlying vector spaces are even dimensional, but in our case the maximal rank condition implies that $\dim \mathfrak{g}/\mathfrak{h}$ is even), so we have

$$\mathcal{S}^+_{L\mathfrak{g}/\mathfrak{t}} - \mathcal{S}^-_{L\mathfrak{g}/\mathfrak{t}} = (\mathcal{S}^+_{L\mathfrak{g}/L\mathfrak{h}} - \mathcal{S}^-_{L\mathfrak{g}/L\mathfrak{h}}) \otimes (\mathcal{S}^+_{L\mathfrak{h}/\mathfrak{t}} - \mathcal{S}^-_{L\mathfrak{h}/\mathfrak{t}}).$$

Using this decomposition to factor the character $\chi(\mathcal{S}^+_{L\mathfrak{g}/\mathfrak{t}} - \mathcal{S}^-_{L\mathfrak{g}/\mathfrak{t}})$, if we apply the Weyl-Kac character formula for $L\mathfrak{g}$ to the left side of (2.22), we obtain

$$\chi\big(\mathcal{H}_{\boldsymbol{\lambda}} \otimes (\mathcal{S}^+_{L\mathfrak{g}/L\mathfrak{h}} - \mathcal{S}^-_{L\mathfrak{g}/L\mathfrak{h}})\big) = \frac{\sum_{w \in \mathcal{W}_{\mathfrak{g}}} (-1)^w e^{w(\boldsymbol{\lambda}+\boldsymbol{\rho}_{\mathfrak{g}})}}{\chi(\mathcal{S}^+_{L\mathfrak{h}/\mathfrak{t}} - \mathcal{S}^-_{L\mathfrak{h}/\mathfrak{t}})}. \tag{2.23}$$

Writing the numerator on the right side of this equation as

$$\sum_{w \in \mathcal{W}_{\mathfrak{g}}} (-1)^w e^{w(\boldsymbol{\lambda}+\boldsymbol{\rho}_{\mathfrak{g}})} = \sum_{c \in \mathcal{C}} (-1)^c \sum_{w \in \mathcal{W}_{\mathfrak{h}}} (-1)^w e^{wc(\boldsymbol{\lambda}+\boldsymbol{\rho}_{\mathfrak{g}})},$$

we can then apply the Weyl-Kac character formula for $L\mathfrak{h}$ to the right side of (2.23), giving us

$$\chi\big(\mathcal{H}_{\boldsymbol{\lambda}} \otimes (\mathcal{S}^+_{L\mathfrak{g}/L\mathfrak{h}} - \mathcal{S}^-_{L\mathfrak{g}/L\mathfrak{h}})\big) = \sum_{c \in \mathcal{C}} (-1)^c \chi\big(\mathcal{U}_{c(\boldsymbol{\lambda}+\boldsymbol{\rho}_{\mathfrak{g}})-\boldsymbol{\rho}_{\mathfrak{h}}}\big),$$

which is the character of the desired identity (2.22). □

EXAMPLE. Take $G = \mathrm{SU}(2)$ and let $H = \mathrm{U}(1)$ be the diagonal elements. The identity (2.22) lets us explicity compute the characters of $\mathcal{S}^{\pm}_{L\mathfrak{g}/L\mathfrak{h}}$. (In general, although the difference $\mathcal{S}^+ - \mathcal{S}^-$ can be computed directly, it is more difficult to determine $\mathcal{S}^{\pm}$ individually.) We observe that $c_G = 2$ and $c_H = 0$, while

$\rho_G = 1$ and $\rho_H = 0$, and the affine Weyl group $\mathcal{W}_H$ is trivial. The lowest weight vector of $\mathcal{S}$ therefore has $S^1 \times \mathrm{SU}(2) \times S^1$ weight $(0, -\rho_G, c_G) = (0, -1, 2)$. The space $\mathcal{S}^+$ (resp. $\mathcal{S}^-$) is then the span of all vectors obtained from this via an even (resp. odd) number of Clifford multiplications by the positive generators E_+ and $E_\pm z^n (n > 0)$ of $L(\mathfrak{g}/\mathfrak{h})$. Since each of these generators shifts the SU(2) weight by ± 2, we see that the SU(2) weights of all elements in $\mathcal{S}^+$ are of the form $4n-1$, while the $\mathcal{S}^-$ weights are all $4n+1$. Hence the weights of $\mathcal{S}^+$ and $\mathcal{S}^-$ are distinct. Applying (2.22), we obtain

$$\mathcal{S}^+ = \sum_{w \in \mathcal{W}_G \text{ even}} \mathcal{U}_{w(0,-1,2)}, \qquad \mathcal{S}^- = \sum_{w \in \mathcal{W}_G \text{ odd}} \mathcal{U}_{w(0,-1,2)}$$

Writing out explicitly the action of the affine Weyl group of SU(2), we have

$$\chi_{\mathcal{S}^+}(z, w, u) = u^2 \sum_{n \in \mathbb{Z}} w^{4n-1} z^{2n^2-n} \prod_{k>0} (1 - z^k)^{-1}$$

$$\chi_{\mathcal{S}^-}(z, w, u) = u^2 \sum_{n \in \mathbb{Z}} w^{4n+1} z^{2n^2+n} \prod_{k>0} (1 - z^k)^{-1}$$

where z corresponds to the energy, w to the SU(2) weight, and u to the level. Here, the level is 2, the sum corresponds to the sum in (2.22), and the product is the character of the irreducible positive energy representation of LS^1.

THEOREM 7. *The $L\mathfrak{g}$-equivariant index of the Dirac operator $\partial\!\!\!/^* \otimes \mathcal{U}_{m,\mu,k}$ on LG/LH twisted by an irreducible positive energy representation $\mathcal{U}_{m,\mu,k}$ of $L\mathfrak{h}$ of lowest weight (m, μ, k) is the virtual $L\mathfrak{g}$ representation*

$$\mathrm{Index}_{L\mathfrak{g}}\, \partial\!\!\!/^* \otimes \mathcal{U}_{m,\mu,k} = (-1)^w \mathcal{H}_{w(m+m_\mathfrak{h}, \mu-\rho_\mathfrak{h}, k+c_\mathfrak{h}) - (m_\mathfrak{g}, -\rho_\mathfrak{g}, c_\mathfrak{g})}$$

if there exists an element $w \in \mathcal{W}_G$ in the affine Weyl group of G such that the weight $w(m + m_\mathfrak{h}, \mu - \rho_\mathfrak{h}, k + c_\mathfrak{h}) - (m_\mathfrak{g}, -\rho_\mathfrak{g}, c_\mathfrak{g})$ is anti-dominant for $L\mathfrak{g}$. If no such w exists, then $\mathrm{Index}_{L\mathfrak{g}}\, \partial\!\!\!/^* \otimes \mathcal{U}_{m,\mu,k} = 0$.

PROOF. We recall from the end of the previous section that

$$\mathrm{Index}_{L\mathfrak{g}}\, \partial\!\!\!/^* \otimes \mathcal{U}_{m,\mu,k} = \sum_{\boldsymbol{\lambda}} \mathcal{H}_{\boldsymbol{\lambda}} \left\langle i^*(\mathcal{H}_{\boldsymbol{\lambda}})(\mathcal{S}^+ - \mathcal{S}^-), \mathcal{U}_{m,\mu,k} \right\rangle_{L\mathfrak{h}}.$$

Applying the homogeneous Weyl-Kac formula (2.22), we then have

$$\begin{aligned}\mathrm{Index}_{L\mathfrak{g}}\, \partial\!\!\!/^* \otimes \mathcal{U}_{m,\mu,k} &= \sum_{\boldsymbol{\lambda}} \mathcal{H}_{\boldsymbol{\lambda}} \Big\langle \sum_{c \in \mathcal{C}} (-1)^c \mathcal{U}_{c(\boldsymbol{\lambda}+\boldsymbol{\rho}_\mathfrak{g}) - \boldsymbol{\rho}_\mathfrak{h}}, \mathcal{U}_{m,\mu,k} \Big\rangle_{L\mathfrak{h}} \\ &= \sum_{\boldsymbol{\lambda}, c} (-1)^c \mathcal{H}_{\boldsymbol{\lambda}}\, \delta_{c(\boldsymbol{\lambda}+\boldsymbol{\rho}_\mathfrak{g}) - \boldsymbol{\rho}_\mathfrak{h}, (m,\mu,k)}.\end{aligned}$$

Finally, noting that

$$c(\boldsymbol{\lambda} + \boldsymbol{\rho}_\mathfrak{g}) - \boldsymbol{\rho}_\mathfrak{h} = (m, \mu, k)$$

if and only if

$$\boldsymbol{\lambda} = c^{-1}\left((m, \mu, k) + \boldsymbol{\rho}_{\mathfrak{h}}\right) - \boldsymbol{\rho}_{\mathfrak{g}},$$

we obtain the desired result for the index. $\square$

Bibliography

[1] A. Alekseev and E. Meinrenken. The Non-Commutative Weil Algebra. Preprint: math.DG/9903052.

[2] R. Bott. The index theorem for homogeneous differential operators. In *Differential and Combinatorial Topology*, S. S. Cairns (ed.), 167–186, Princeton University Press, 1965.

[3] B. Gross, B. Kostant, P. Ramond, S. Sternberg. The Weyl character formula, the half-spin representations, and equal rank subgroups. *Proc. Natl. Acad. Sci. USA* 95, 8441–8442, 1998.

[4] B. Kostant. A cubic Dirac operator and the emergence of Euler number multiplets of representations for equal rank subgroups. *Duke Math. Journal*, to appear.

[5] D. A. McLaughlin. Orientation and string structures on loop space. *Pacific J. Math.* 155, 143-156, 1992.

[6] A. Pressley and G. Segal. *Loop Groups*. Oxford University Press, 1988.

[7] C. H. Taubes. S^1 actions and elliptic genera. *Commun. Math. Phys.* 122, 455–526, 1989.

[8] C. Teleman. Lie algebra cohomology and the Fusion Rules. *Commun. Math. Phys.* 173, 265–311, 1995.

[9] E. Witten. The index of the Dirac operator in loop space. In *Lecture Notes in Mathematics* 1326, P. S. Landweber (ed.), 161–181, Springer-Verlag, 1988.